



Engineering Drawing (EDRG101)

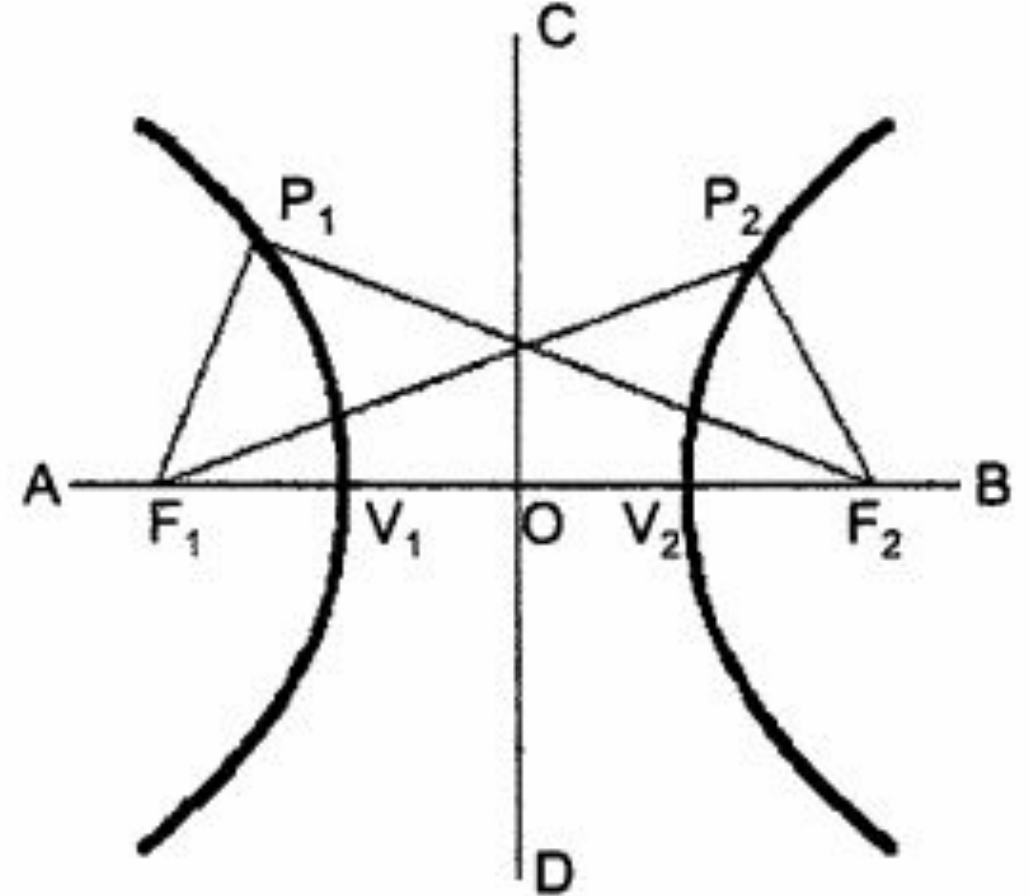
Lecture 4: Engineering Curves

Hyperbola:-

Hyperbola is the curve generated by a point moving so that the difference between its distances from two fixed points F_1 and F_2 is constant and equal to the distances between vertices.

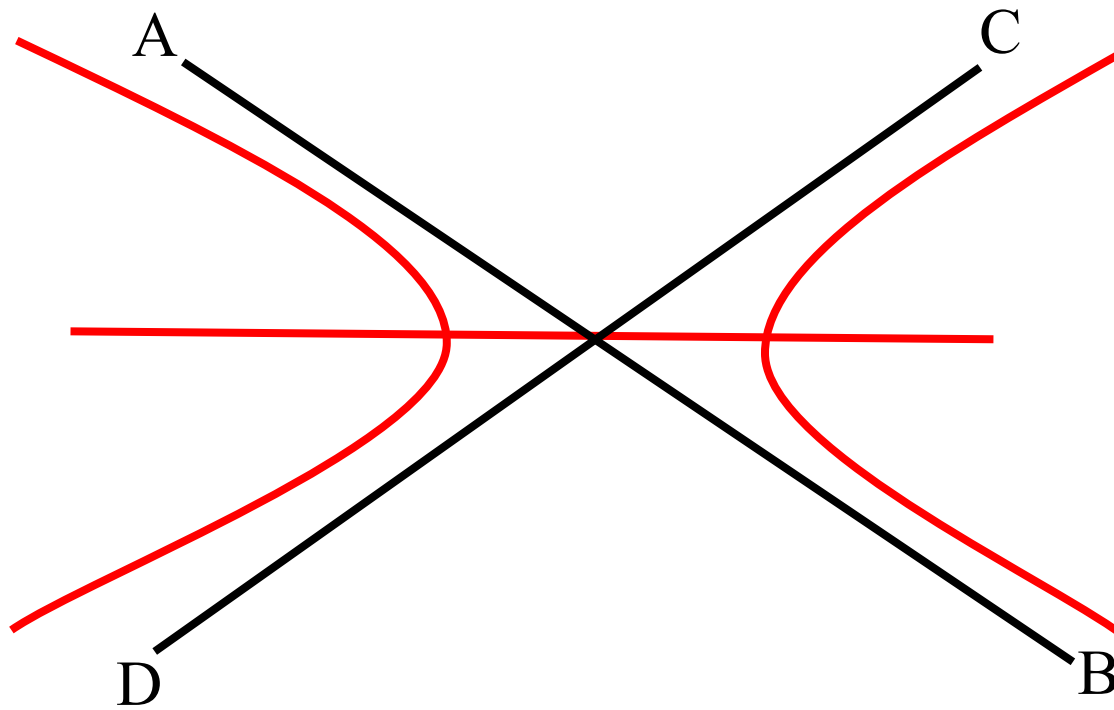
$$\text{i.e. } PF_1 - PF_2 = V_1V_2$$

Here V_1V_2 is called transverse axis.



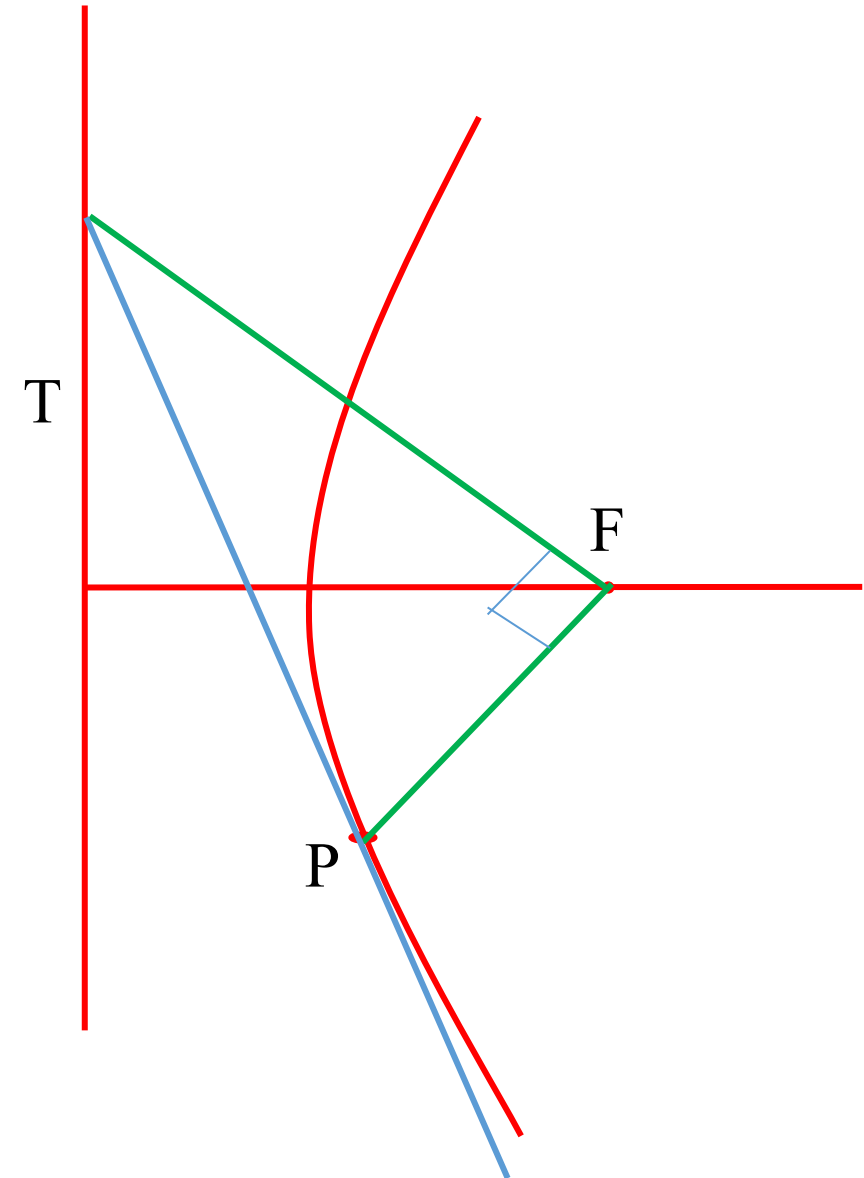
Asymptotes

- Asymptotes are the lines which are tangent to the curve at infinity.



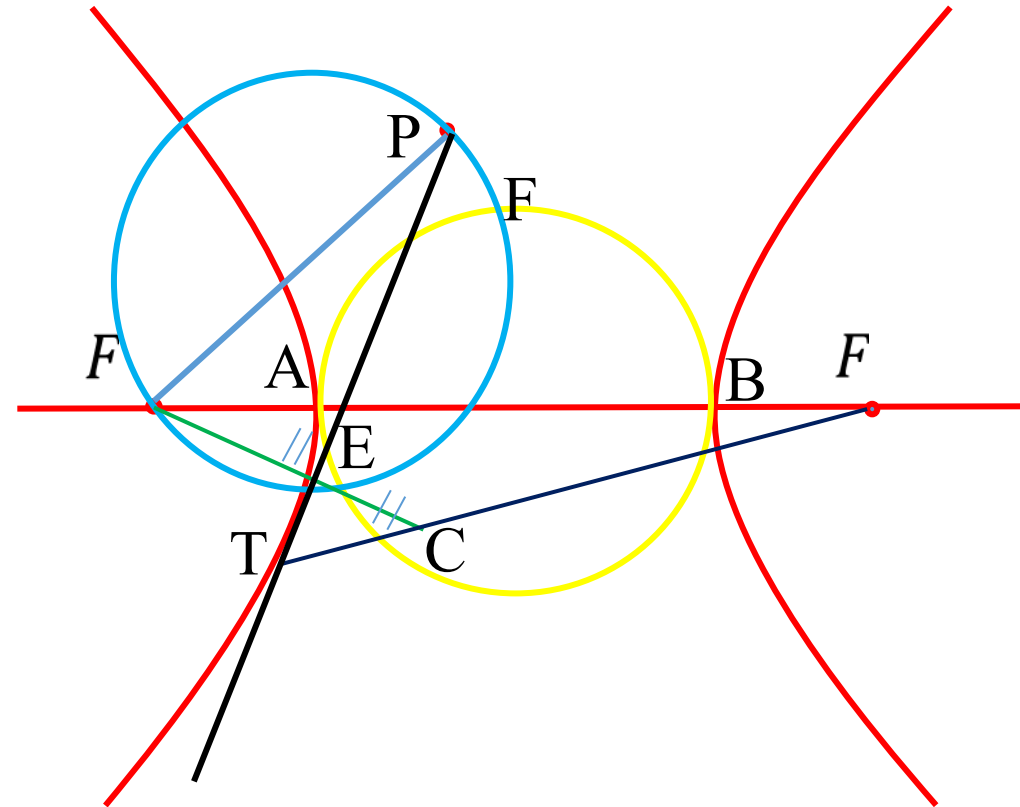
Tangent at a point on hyperbola

- Construct the hyperbola where F be the focus and P is the point where we want to draw tangent.
- Join P to F.
- Draw another line perpendicular to PF which cut directrix at point T.
- Join T and P. Here TP is required tangent to hyperbola



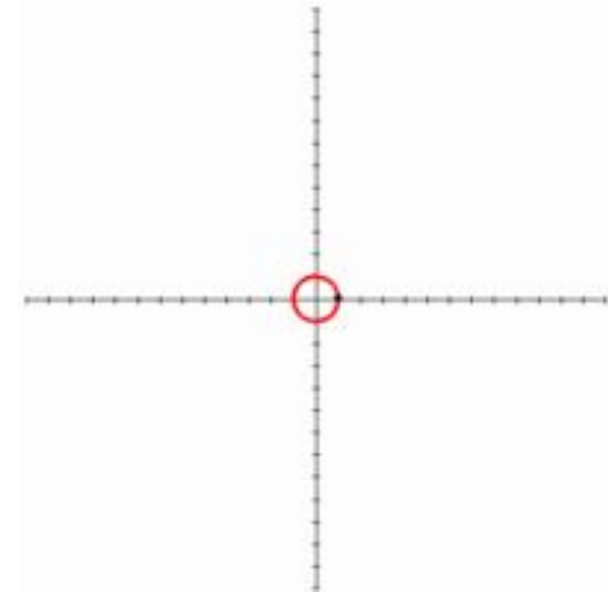
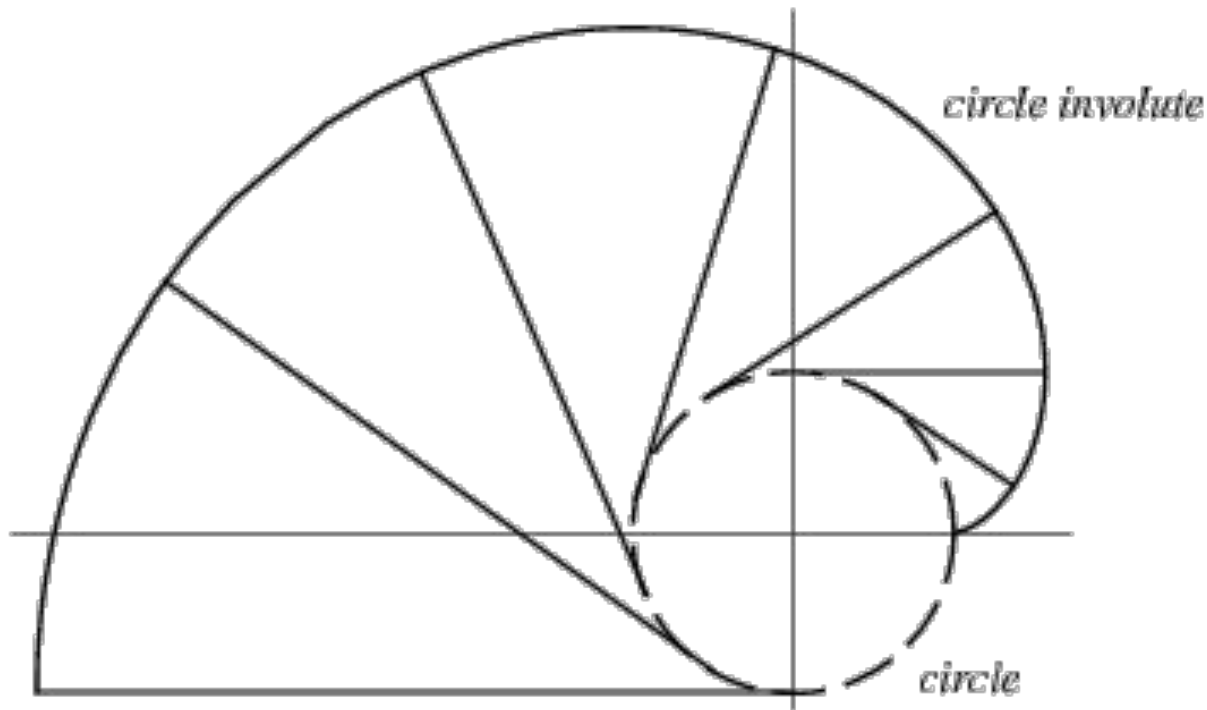
Tangent at a point outside hyperbola

- Draw hyperbola and show its foci F_1 and F_2 and vertices A and B .
- Locate the point P from where tangent to be drawn.
- Join P to nearest focus.
- Draw a circle with PF_1 as diameter.
- Draw another circle with AB as diameter.
- These two circles intersect at E and F .
- Join F_1 to E and produce to C such that $F_1E = EC$.
- Join C and F_2 produce it which cut the curve at T .
- Then join T to P . line TP is required tangent.



Involute

- An *involute* is a curve traced by the free end of a thread unwound from a circle or a polygon, in such a way that the thread is always tight and tangential to the circle or the sides of the polygon.
- Depending on whether the involute is traced over a circle or a polygon, the involute is called an *involute of circle* or *involute of polygon*



Spiral

- **Spiral** is the locus of a point which moves around a circle, called pole which moving away from center.
- Two types
 - Archimedean spiral
 - Logarithmic spiral

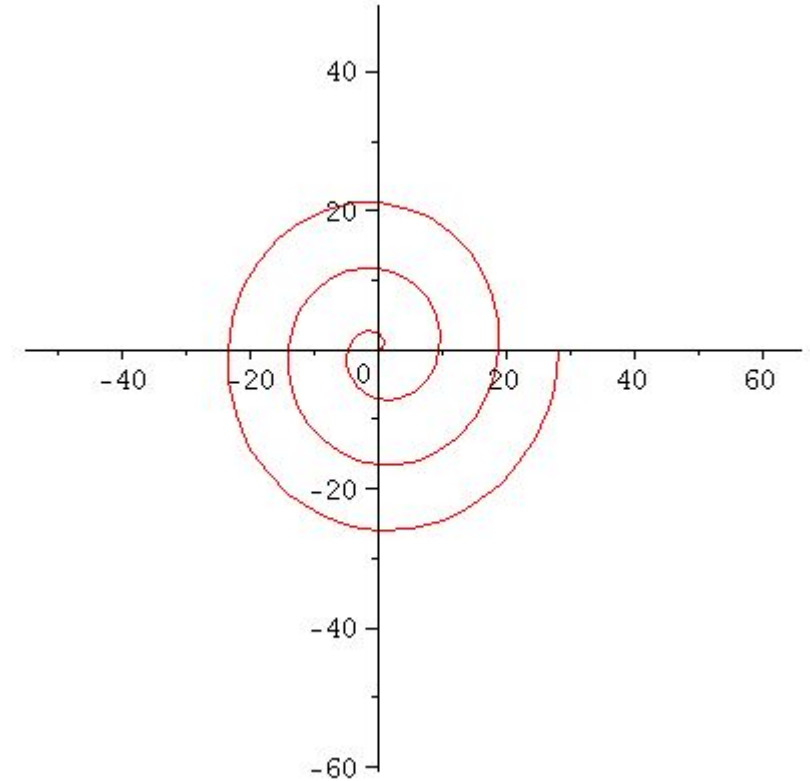
- **Archimedean Spiral;**

Polar equation of Archimidean Spiral is $r = c + a\theta$;
where r =radius vector,

c = initial radius vector

θ =corresponding vectorial in

radian



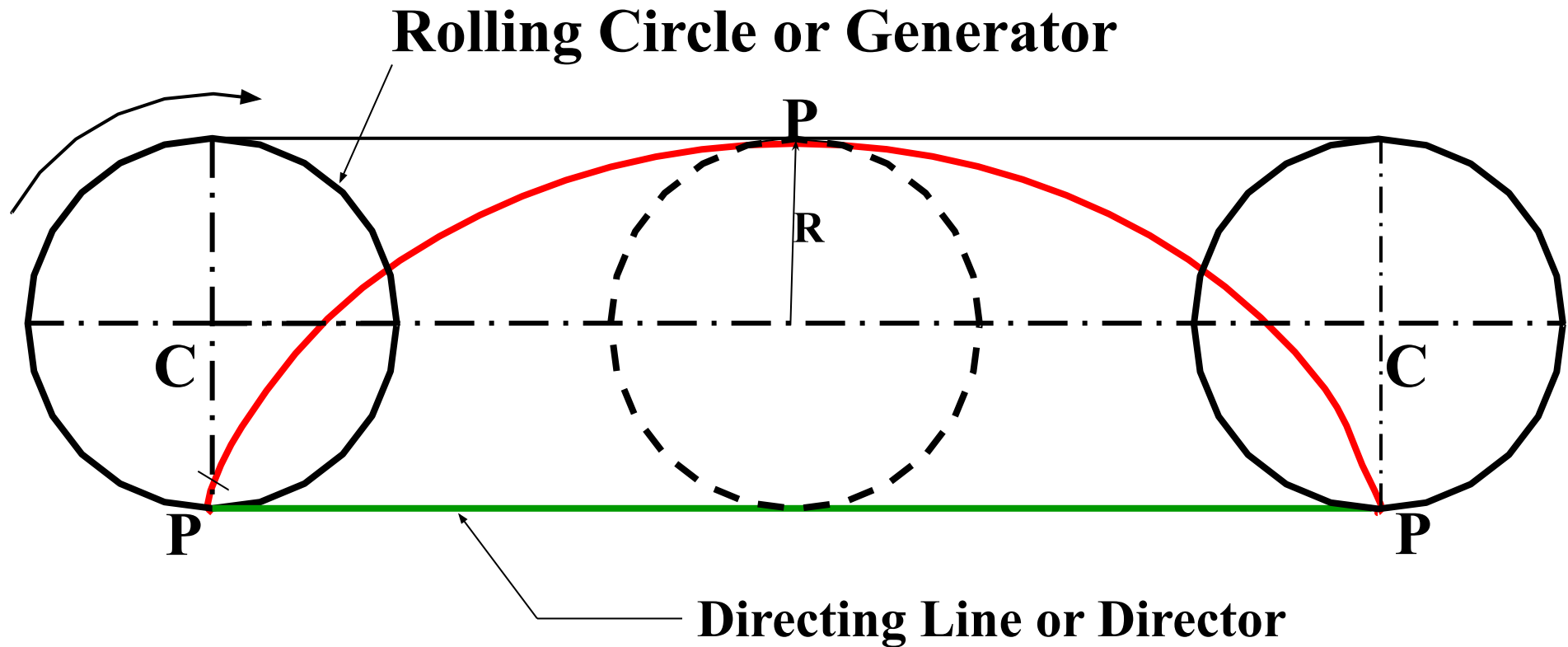
Cyclodial Curves

- When one curve rolls over another curve without slipping or sliding, the path Of any point of the rolling curve is called as **ROULETTE**.
- When rolling curve is a circle and the curve on which it rolls is a straight line Or a circle, we get **CYCLOIDAL GROUP OF CURVES**



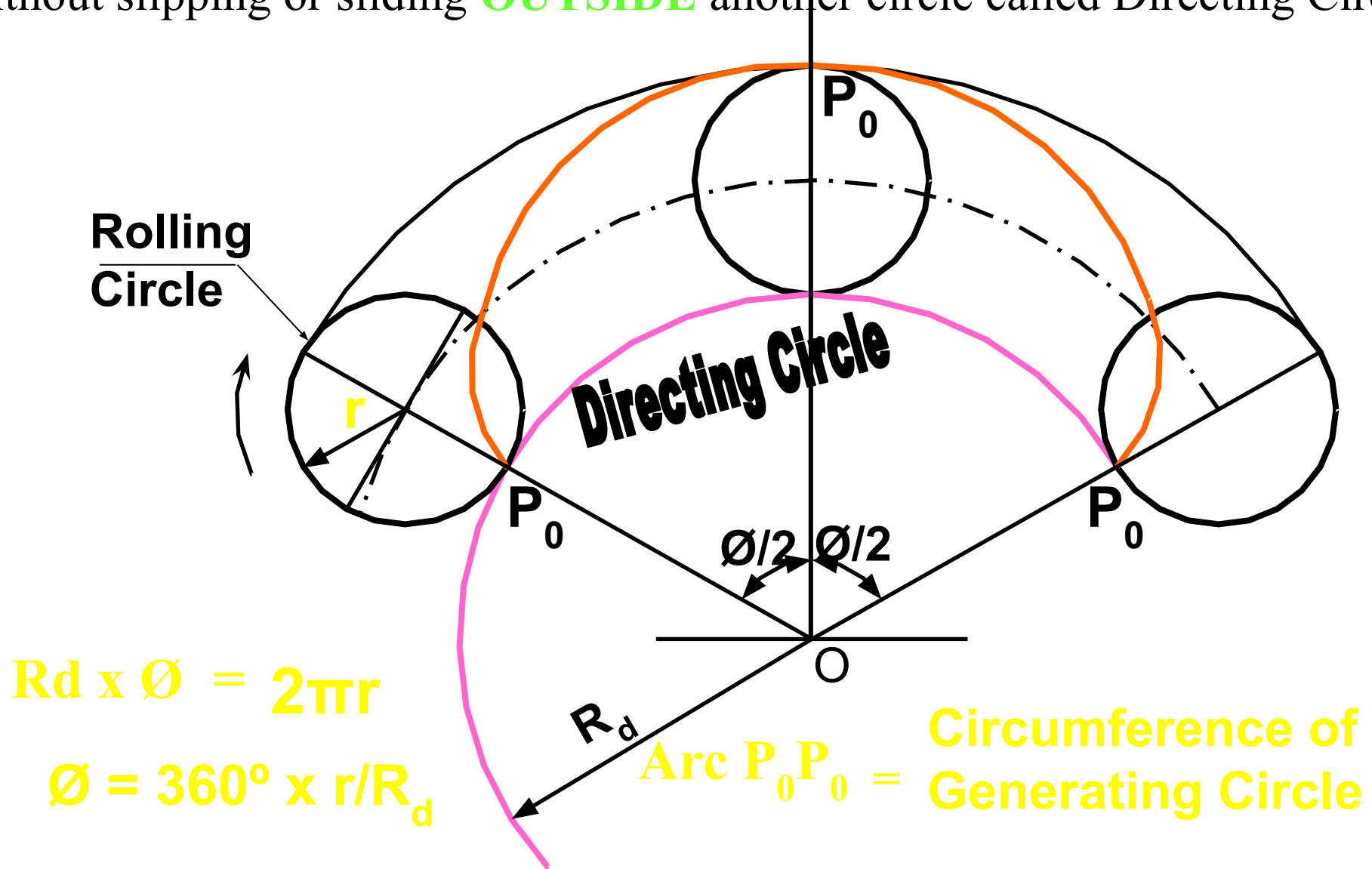
CYCLOID:-

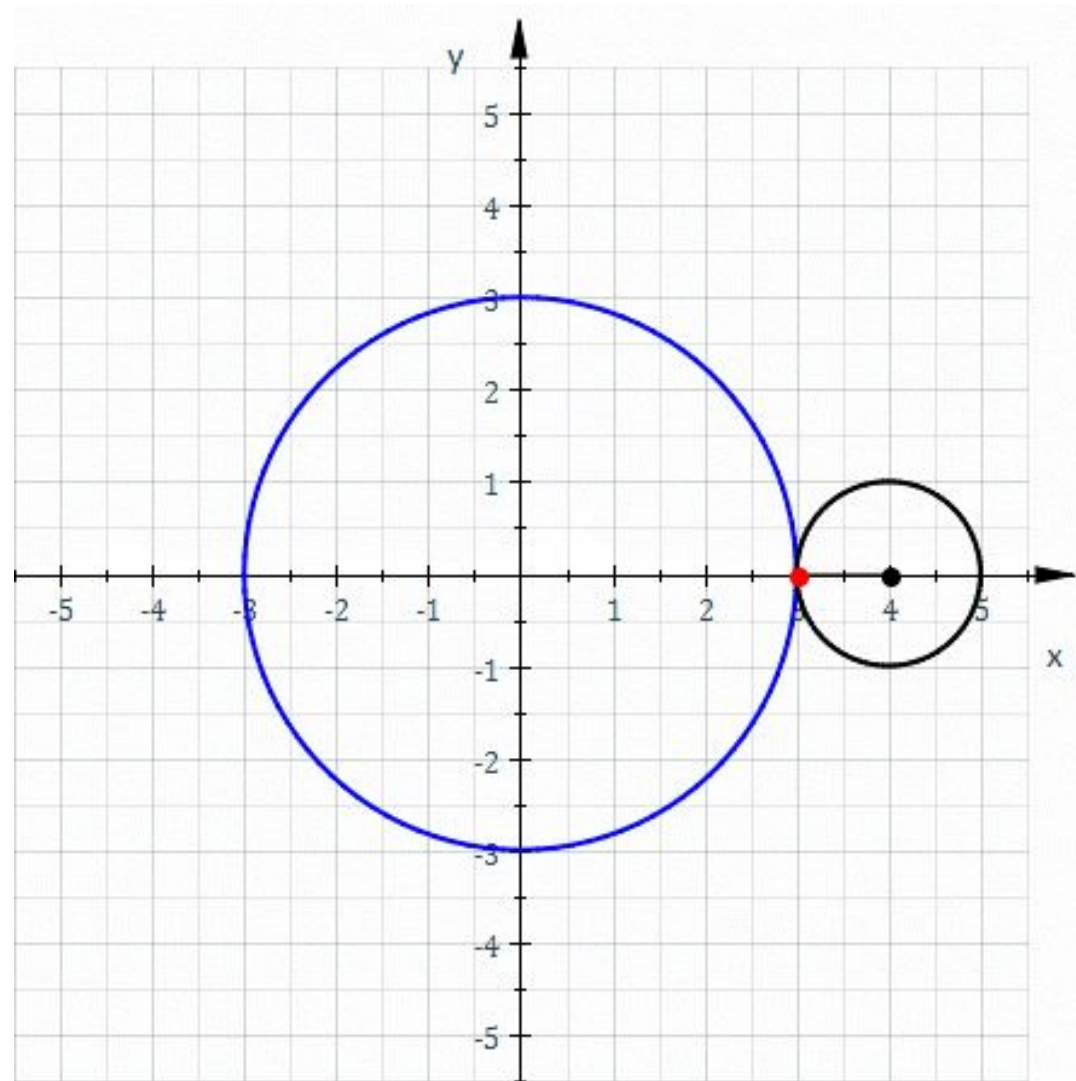
Cycloid is a locus of a point on the circumference of a rolling circle(generator), which rolls without slipping or sliding along a fixed straight line or a directing line or a director.



EPICYCLOID:-

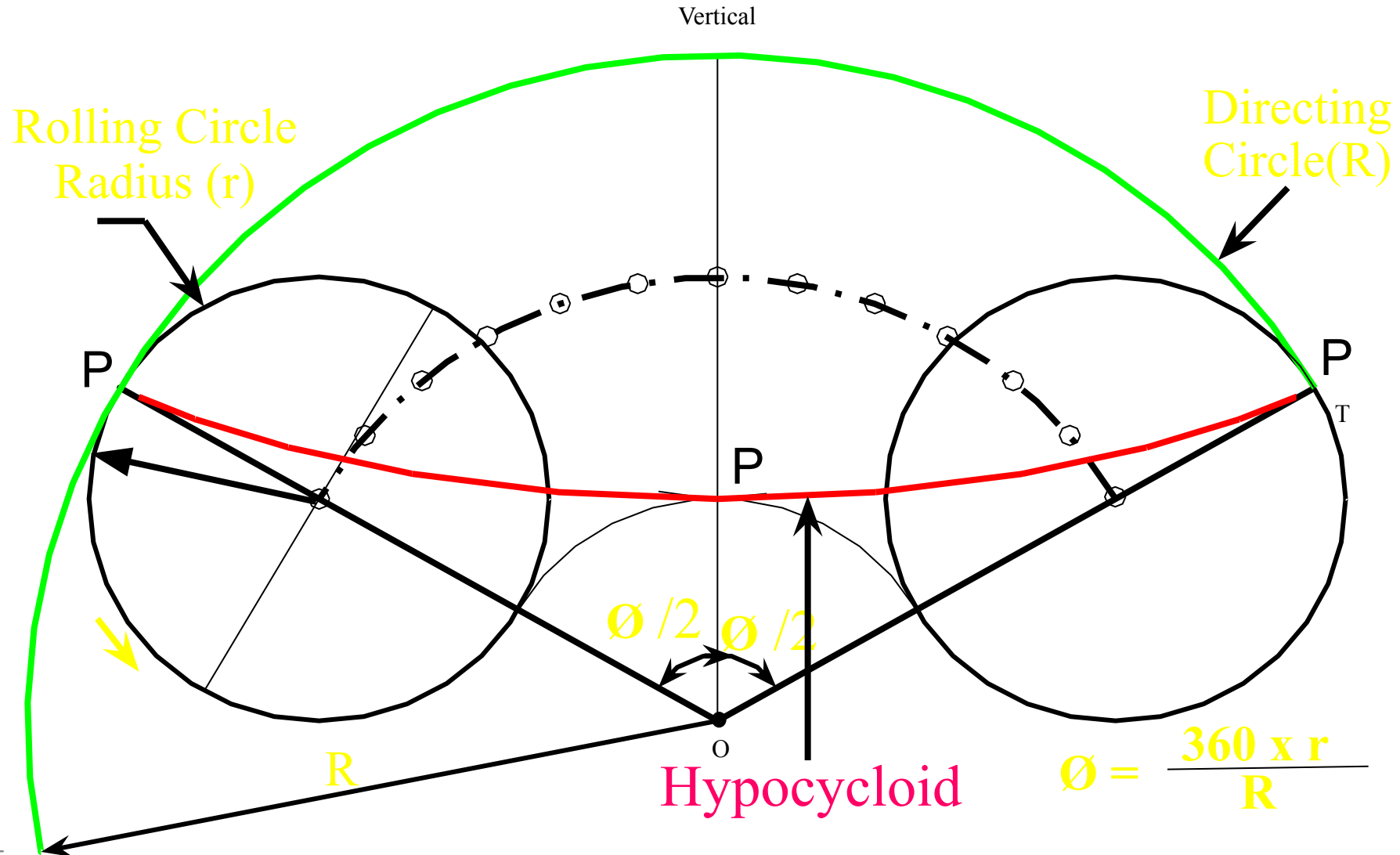
Epicycloid is a locus of a point(P) on the circumference of a rolling circle(generator), which rolls without slipping or sliding **OUTSIDE** another circle called Directing Circle.

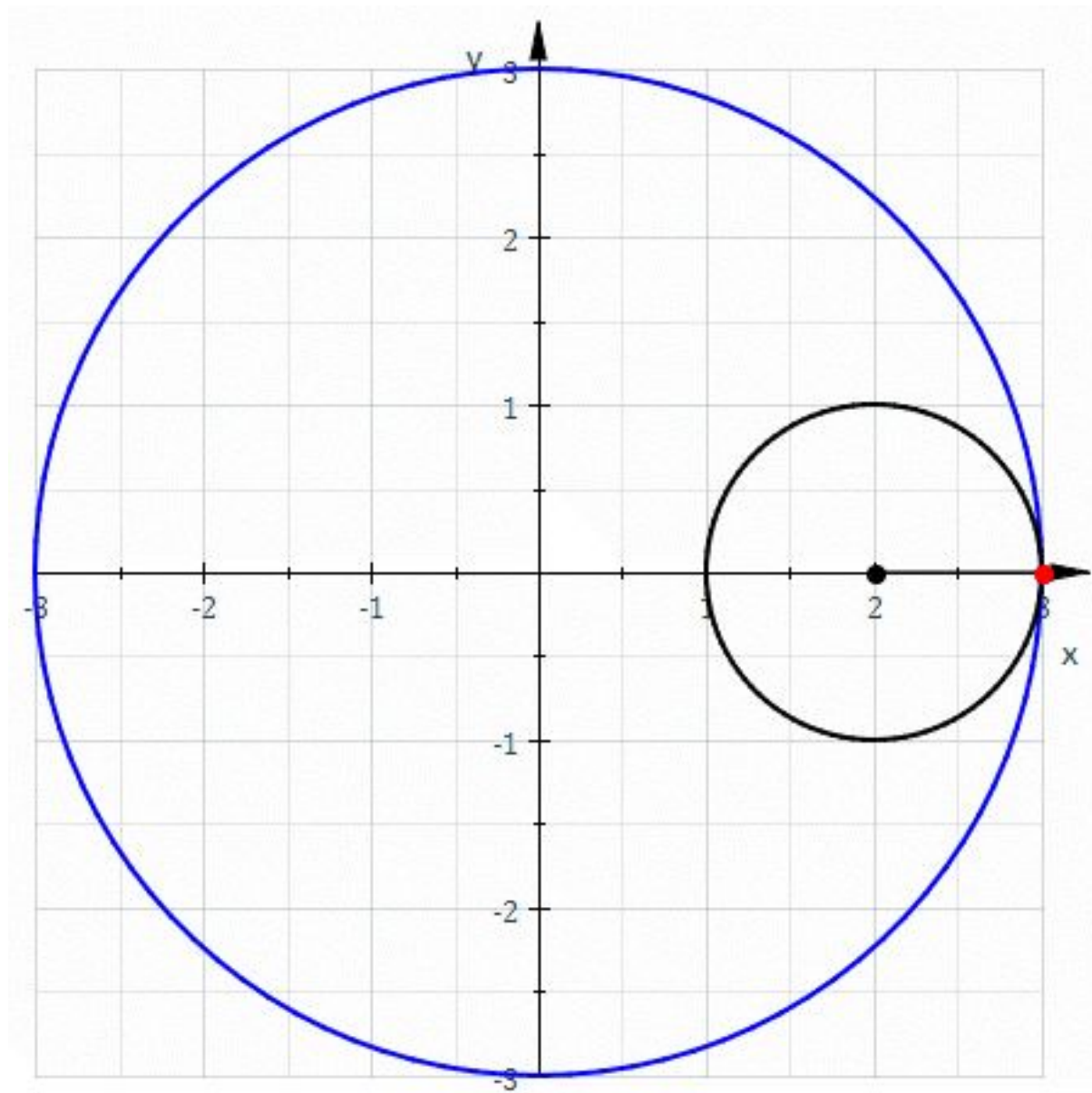




HYPOCYCLOID:-

Hypocycloid is a locus of a point(P) on the circumference of a rolling circle(generator), which rolls without slipping or sliding **INSIDE** another circle called Directing Circle.'





TROCHOID

Trochoid is a locus of a point **inside/outside** the circumference of a rolling circle, which rolls without slipping or sliding along a fixed straight line or a fixed circle.

If the point is **inside** the circumference of the circle, it is called **inferior trochoid**.

If the point is **outside** the circumference of the circle, it is called **superior trochoid**.

Exercise 5-1 (Hyperbola)

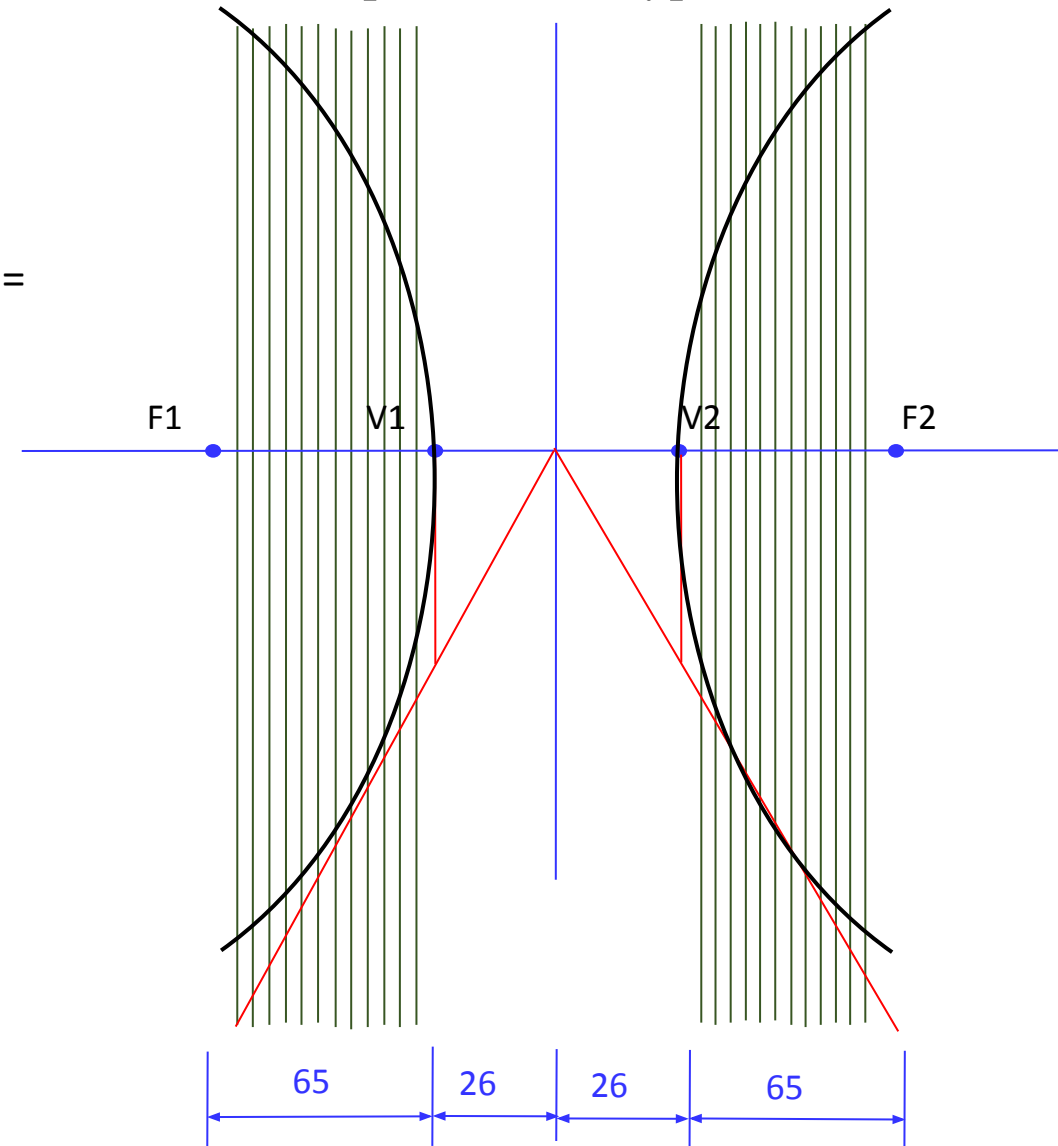
The vertex of a hyperbola is 65mm from its focus. Draw the two parts of the hyperbola if eccentricity is 2.5.

$$e = \frac{\text{Distance of a point from focus}}{\text{Distance of the point from directrix}}$$

Distance of the point from directrix

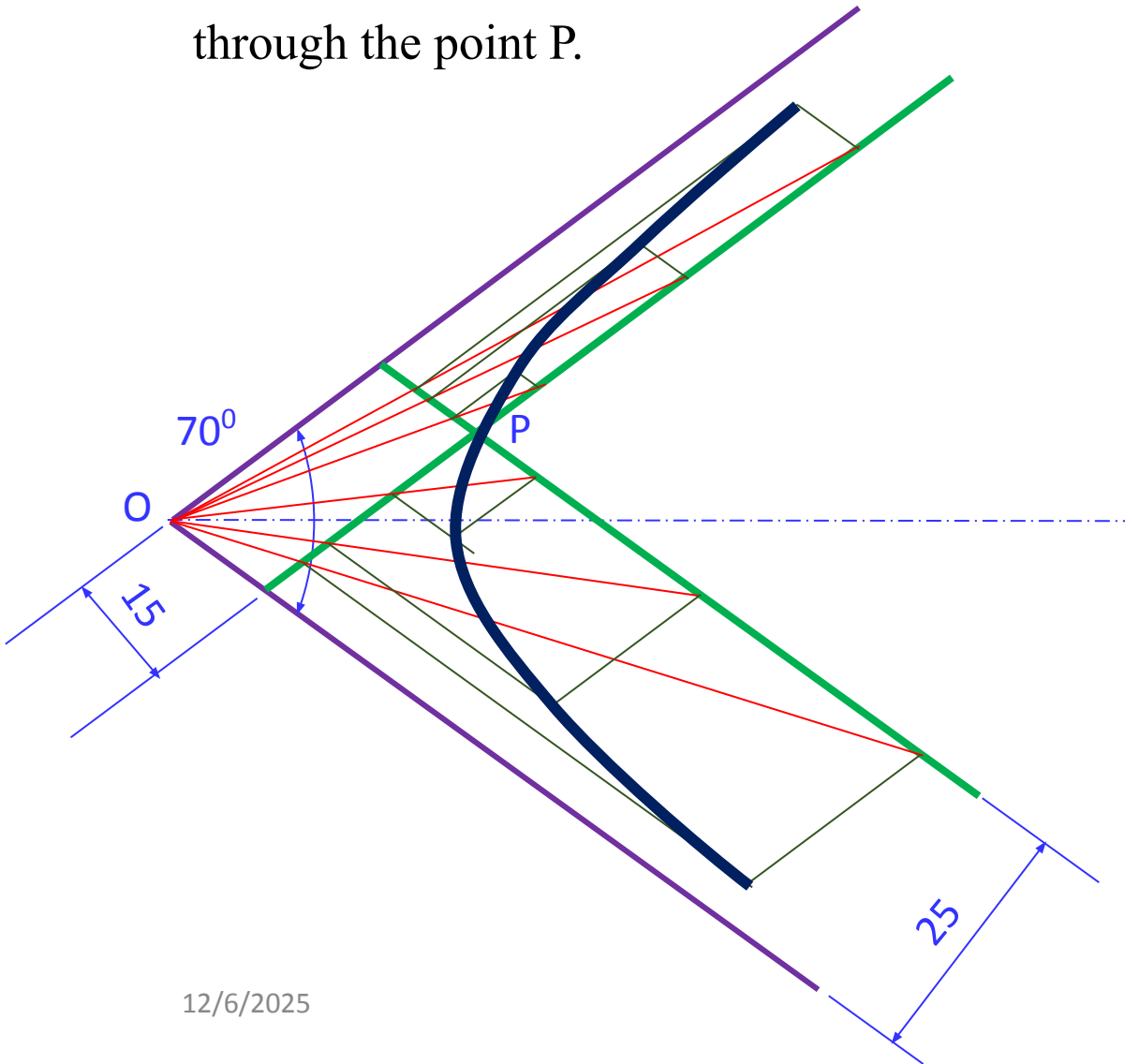
$$\text{Distance of the point (vertex) from directrix} = 65/2.5 = 26 \text{ mm}$$

1. Locate focal points and vertices.
2. Draw governor with the help of eccentricity.
3. Draw hyperbola as shown in previous slide.



Exercise 5-2 (Hyperbola)

The asymptotes of a hyperbola are inclined at 70 degree to each other. A point P on the curve is at a distance of 25mm and 15mm from the two asymptotes. Draw two parts of the hyperbola passing through the point P.

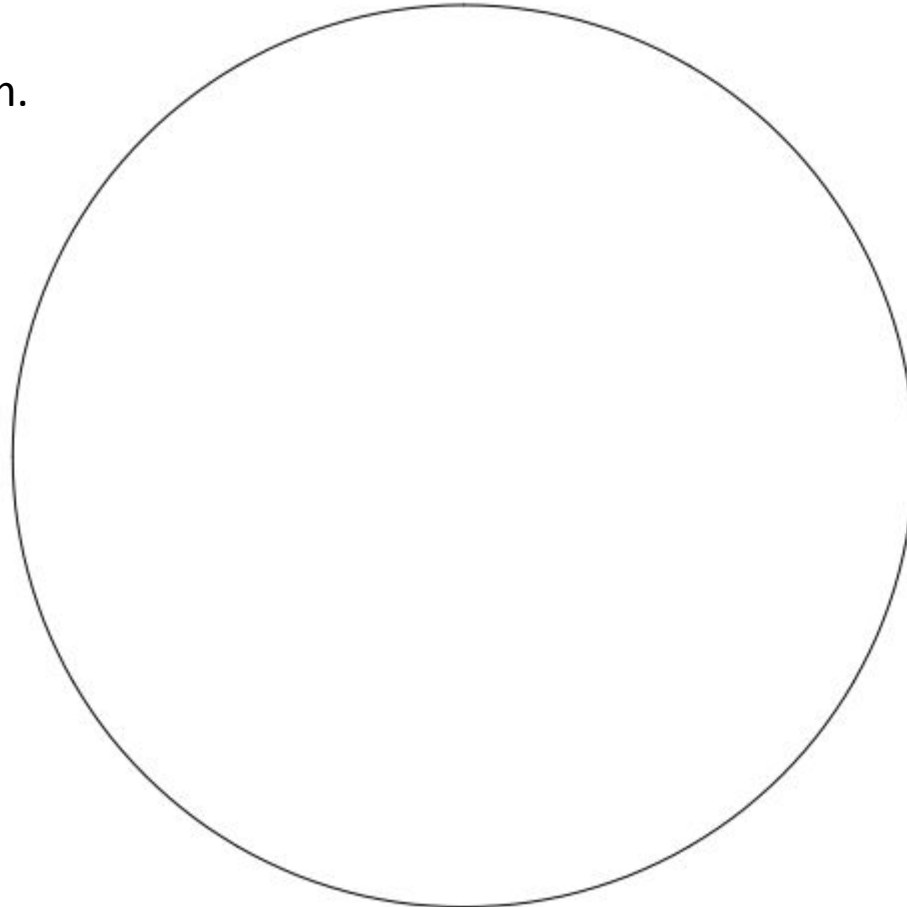


1. Draw asymptotes at an angle of 70 degree.
2. Draw two lines parallel to these asymptotes one at a distance of 25 mm and the other at a distance of 15 mm.
3. Draw random lines from O to meet these newly formed lines as shown.
4. Now through the intersections draw lines parallel to both the asymptotes as shown.
5. Connect the intersections points as shown.

Exercise 5-3 (Involute of a circle)

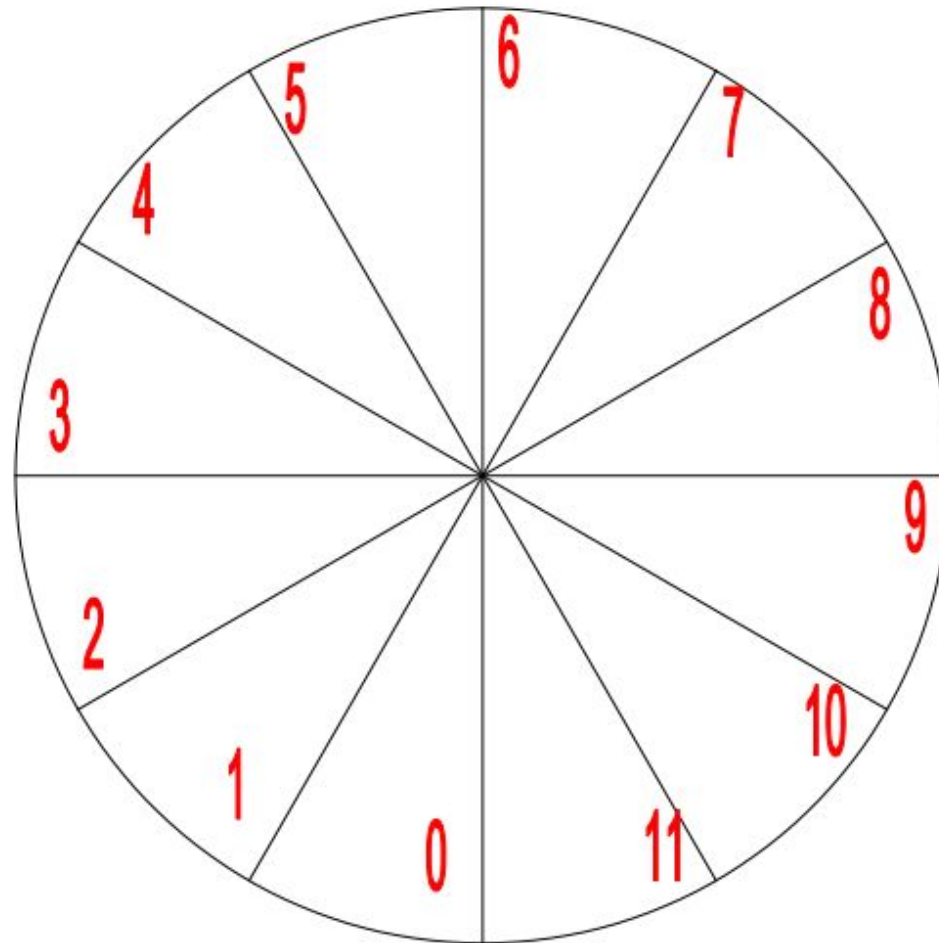
Draw the locus of the end point of a cable unwinding itself from a drum of 80mm diameter such that the unwound cable is always taut.

1. Draw a circle with diameter of 80 mm.



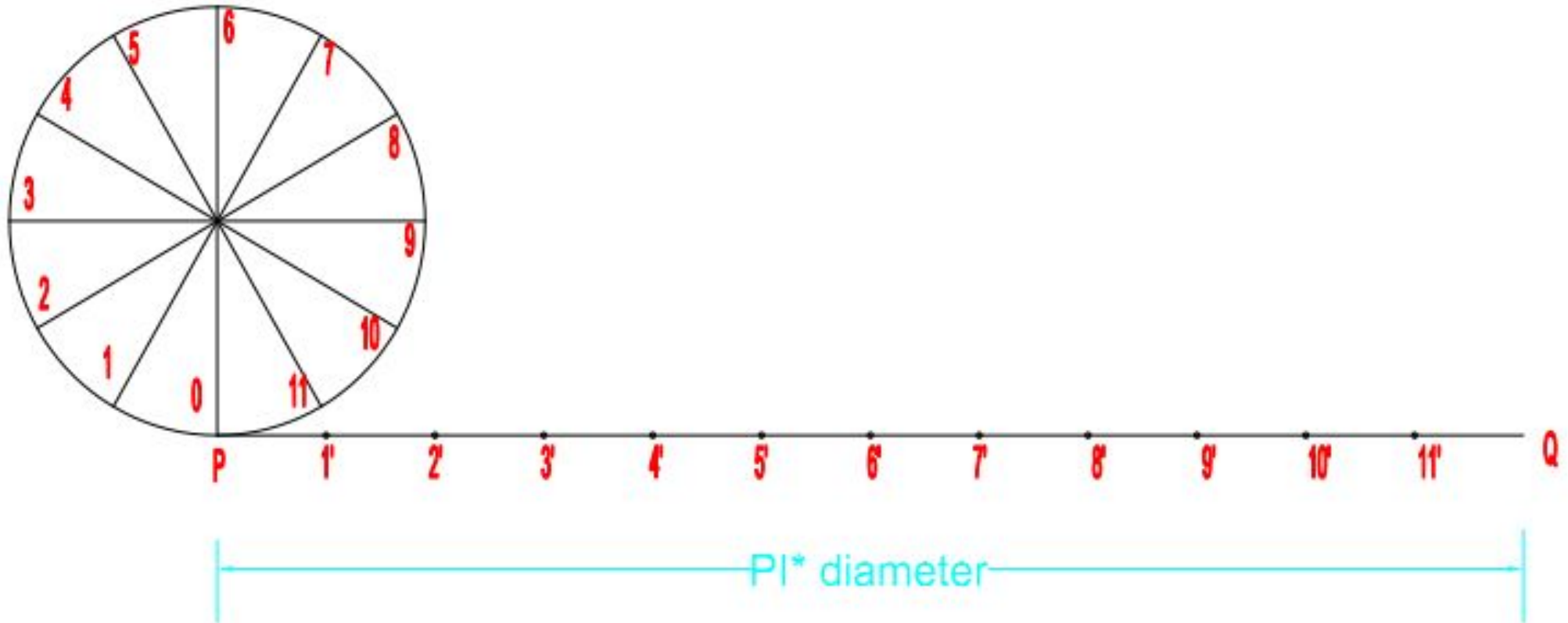
Exercise 5-3 (Involute of a circle)

2. Divide it into 12 equal parts.

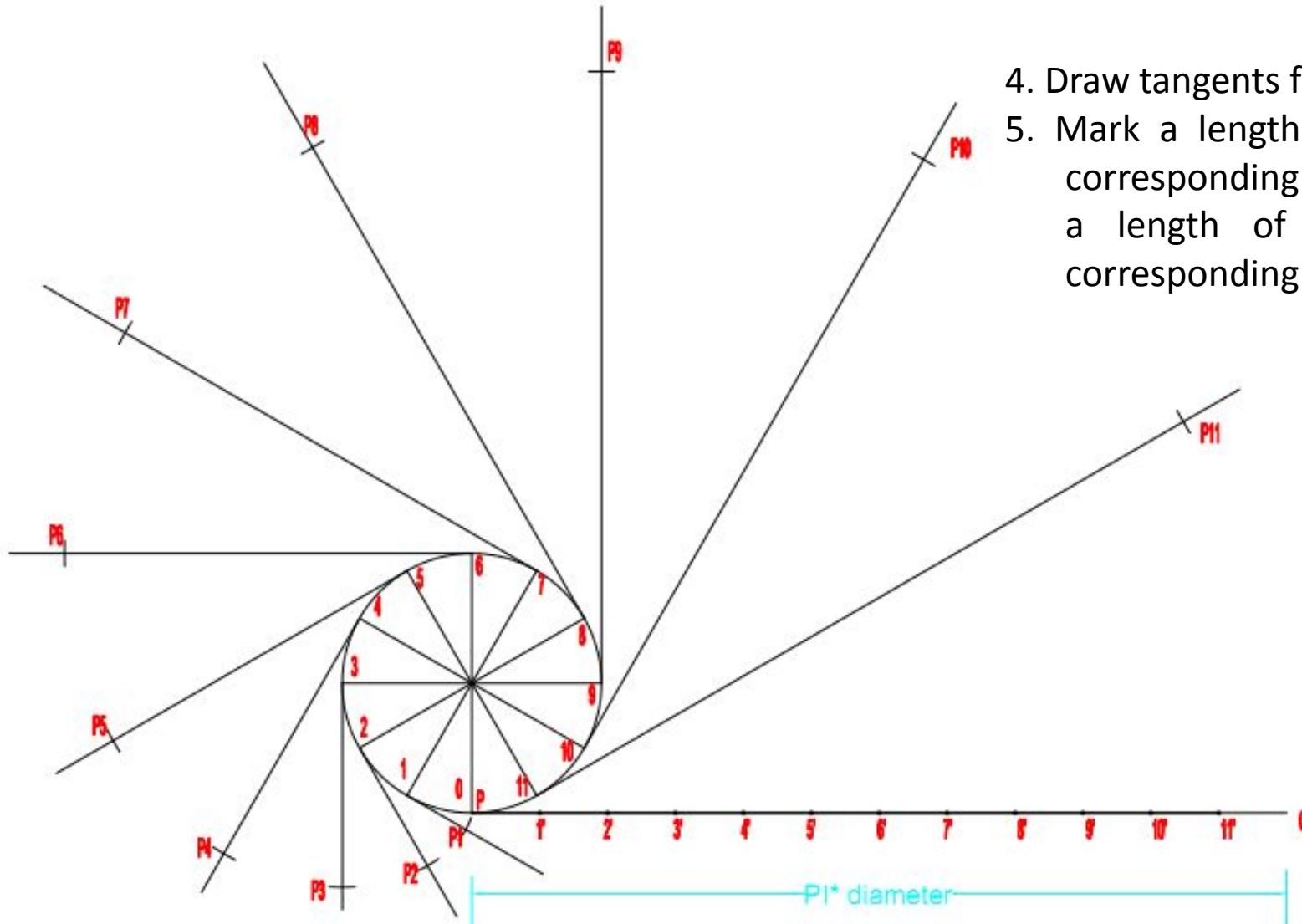


Exercise 5-3 (Involute of a circle)

3. Draw a line tangent to this circle with length equal to circumference of the circle, and divide it into 12 equal parts.

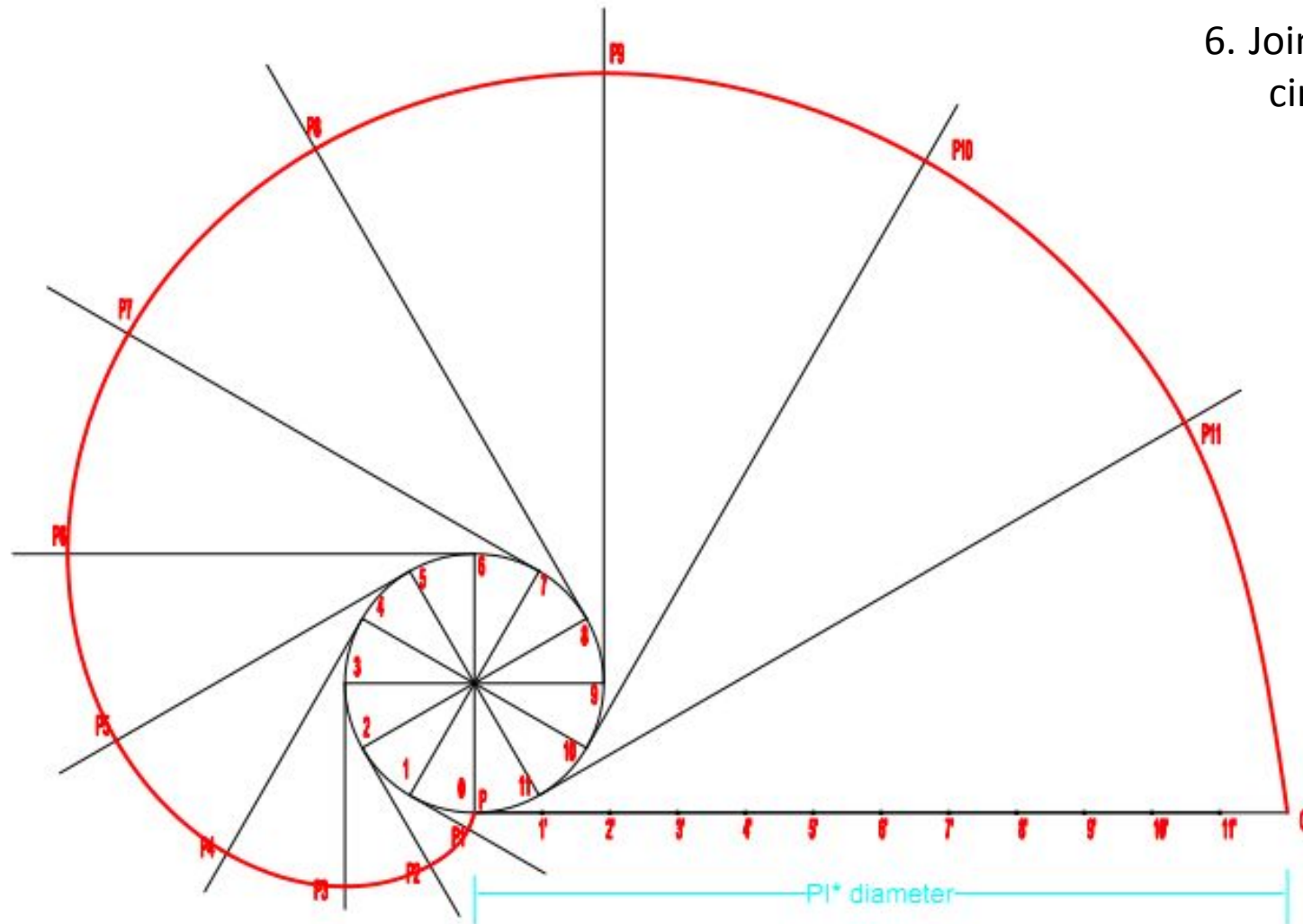


Exercise 5-3 (Involute of a circle)



4. Draw tangents from points 0-11.
5. Mark a length of $P1'$ from 1 on the corresponding tangent. Similarly mark a length of $P2'$ from 2 on the corresponding tangent. And so on.

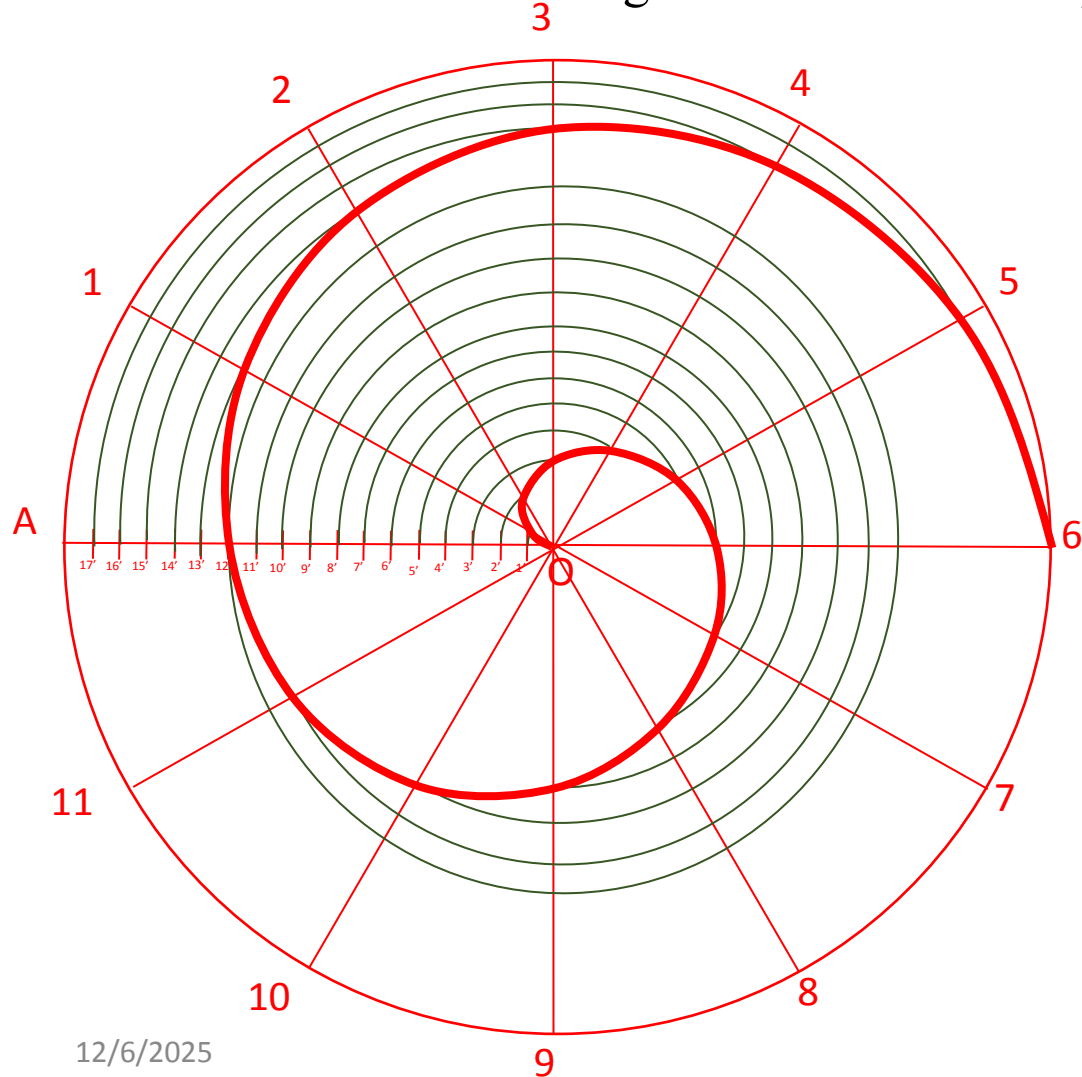
Exercise 5-3 (Involute of a circle)



6. Join the points to form an involute of a circle.

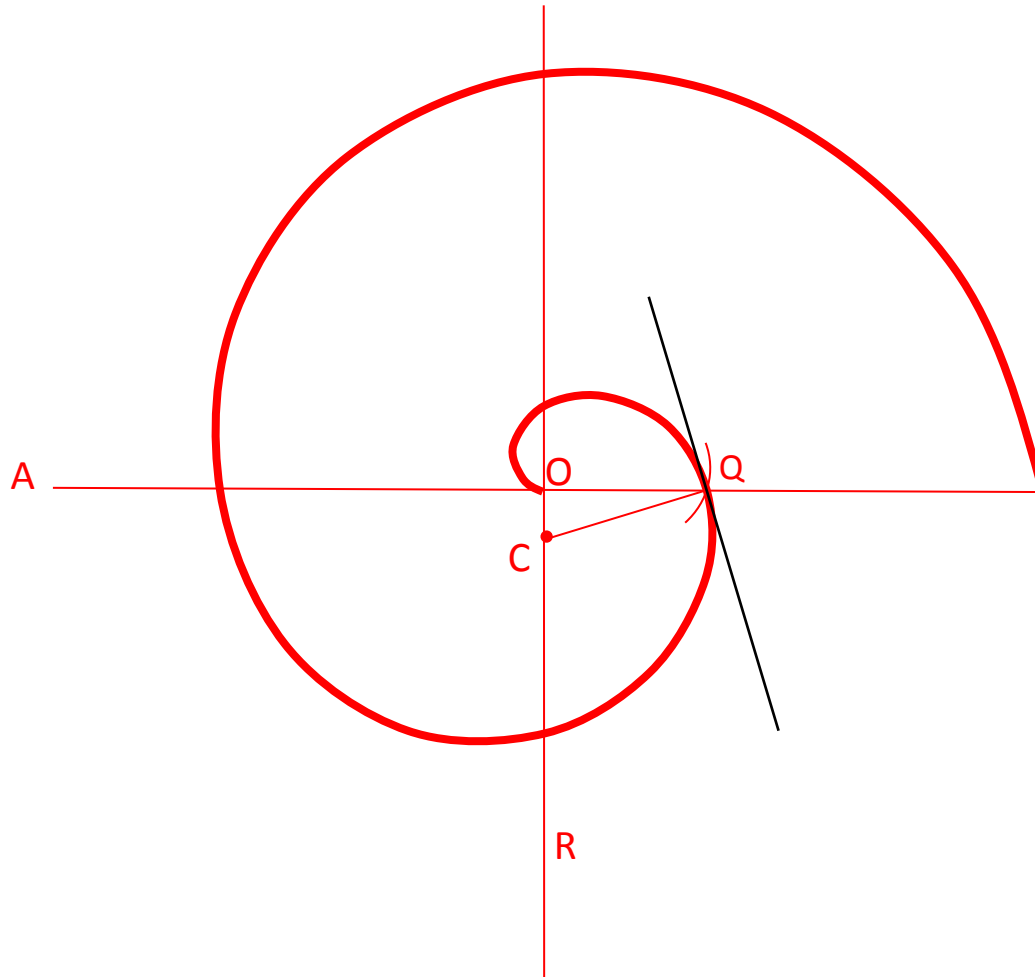
Exercise 5-4 (Archimedian Spiral)

Draw the Archimedean spiral for one and half convolution. The spiral starts from the pole and the greatest radius is 75mm. draw the tangent to the curve at a point 25mm from the pole.



1. Draw a circle of radius 75 mm.
2. Divide it into 12 equal parts.
3. Divide a line into 18 equal parts (12 parts for one convolution, 18 parts for 1 and half convolution)
4. Draw an arc of radius $O1'$ from O upto division 1 of circle. Draw another arc of radius $O2'$ from O upto division 2 of circle. And so on.
5. Join the end points of arcs to get Archimedean Spiral.

Exercise 5-4 (Archimedean Spiral)



Polar equation for Archimedean Spiral : $r = c + a\theta$

$r = 25 \text{ mm}$

$c = 0$ (Because spiral starts from pole)

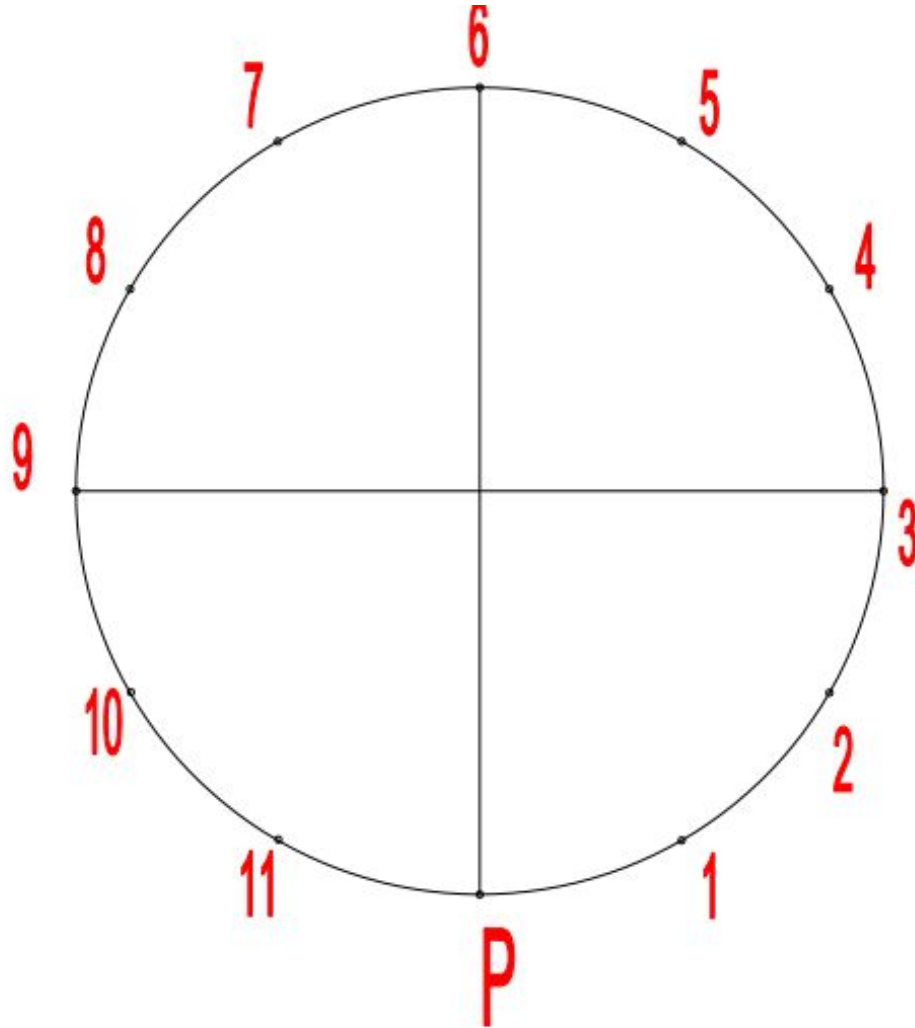
$\theta = 180^\circ = \pi \text{ radian} = 3.14$ (Take angle along the direction of spiral.)

Then, $a = 25/3.14 = 7.96 \text{ mm} \sim 8 \text{ mm}$

1. To make the tangent, mark a length of 25 mm from O on the spiral (Q).
2. Join Q to O.
3. Draw a line (OR) perpendicular to OQ.
4. From polar equation of Archimedean Spiral, find out the value of a.
5. Mark a point C on OR at a length equal to a.
6. Join C and Q.
7. Draw a line perpendicular to CQ through Q which is the required tangent.

Exercise 5-5

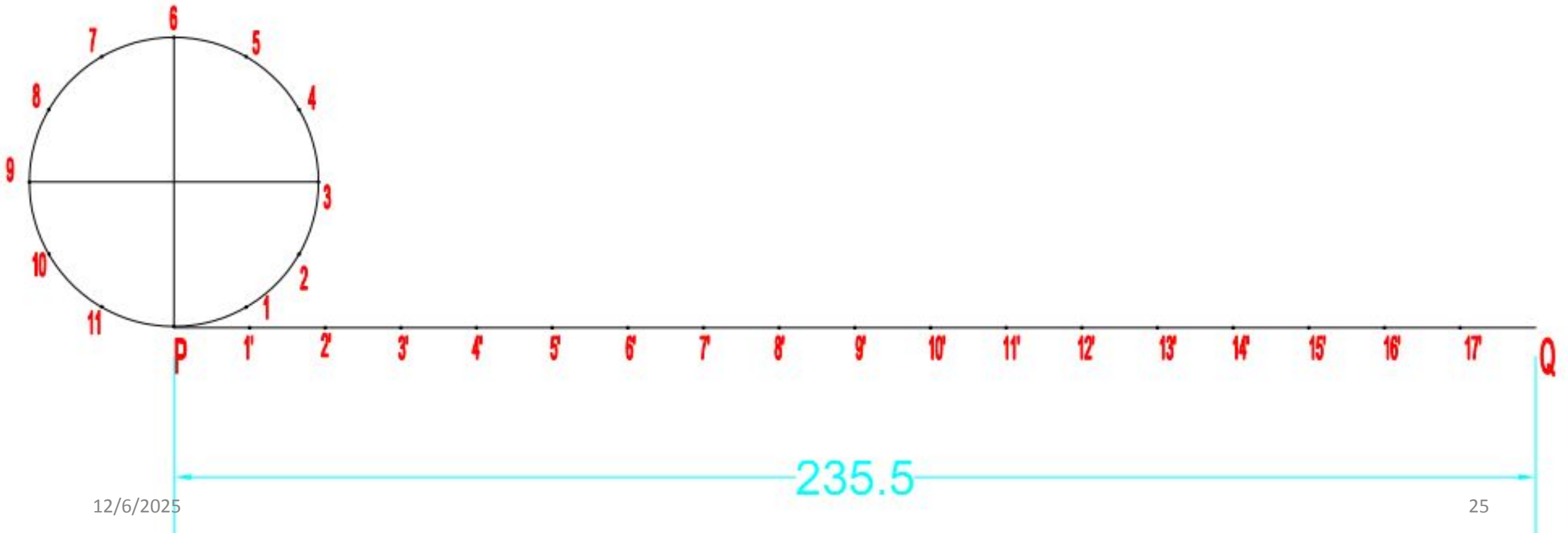
A circle of 50mm diameter rolls on a straight line without slipping. Trace the locus of a point on the circumference if the circle rolls for one and half revolution. Name the curve. Draw tangent and normal to the curve at a point 35mm above the straight line and on the ascending side of the curve.



1. Draw a circle of diameter 50 mm and divide it into 12 parts.

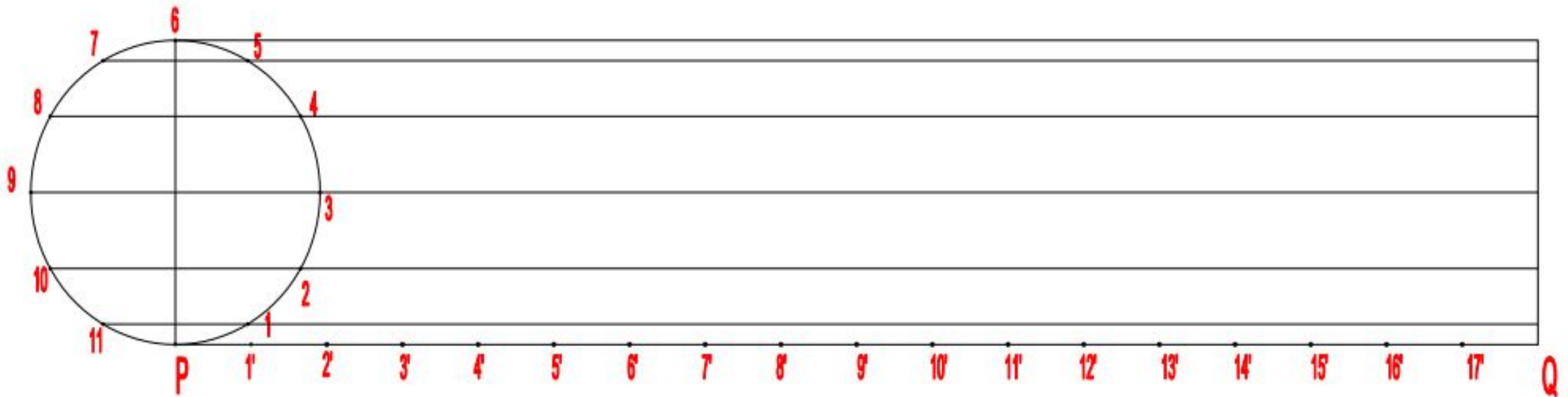
Exercise 5-5

2. Draw a line tangent to the circle of length equal to 1.5 times the circumference. (1 circumference for 1 revolution, 1.5 circumference for 1 and half revolution).
3. Divide it into 18 equal parts. (12 parts for 1 revolution, 18 parts for 1 and half revolution).



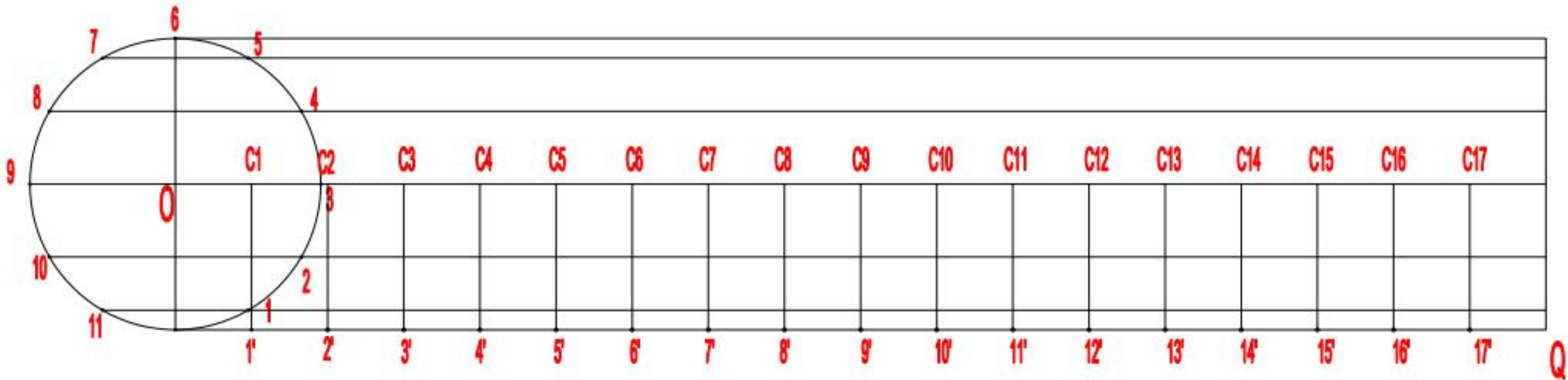
Exercise 5-5

4. Extend the dividing points of the circle horizontally as shown.



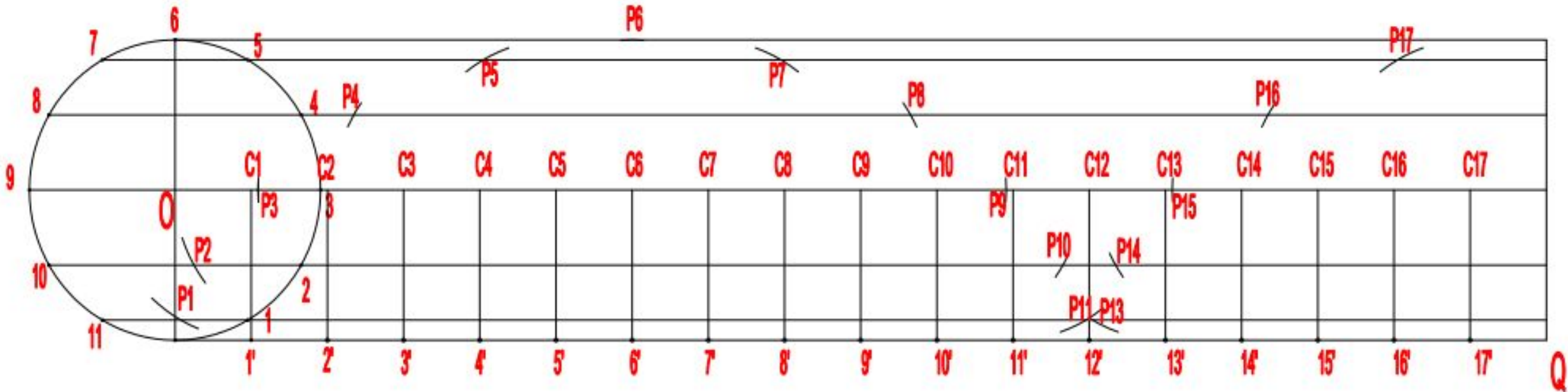
Exercise 5-5

5. Extend the dividing points of the tangent vertically as shown to get the locus of the circle (C1, C2, C3, etc.)



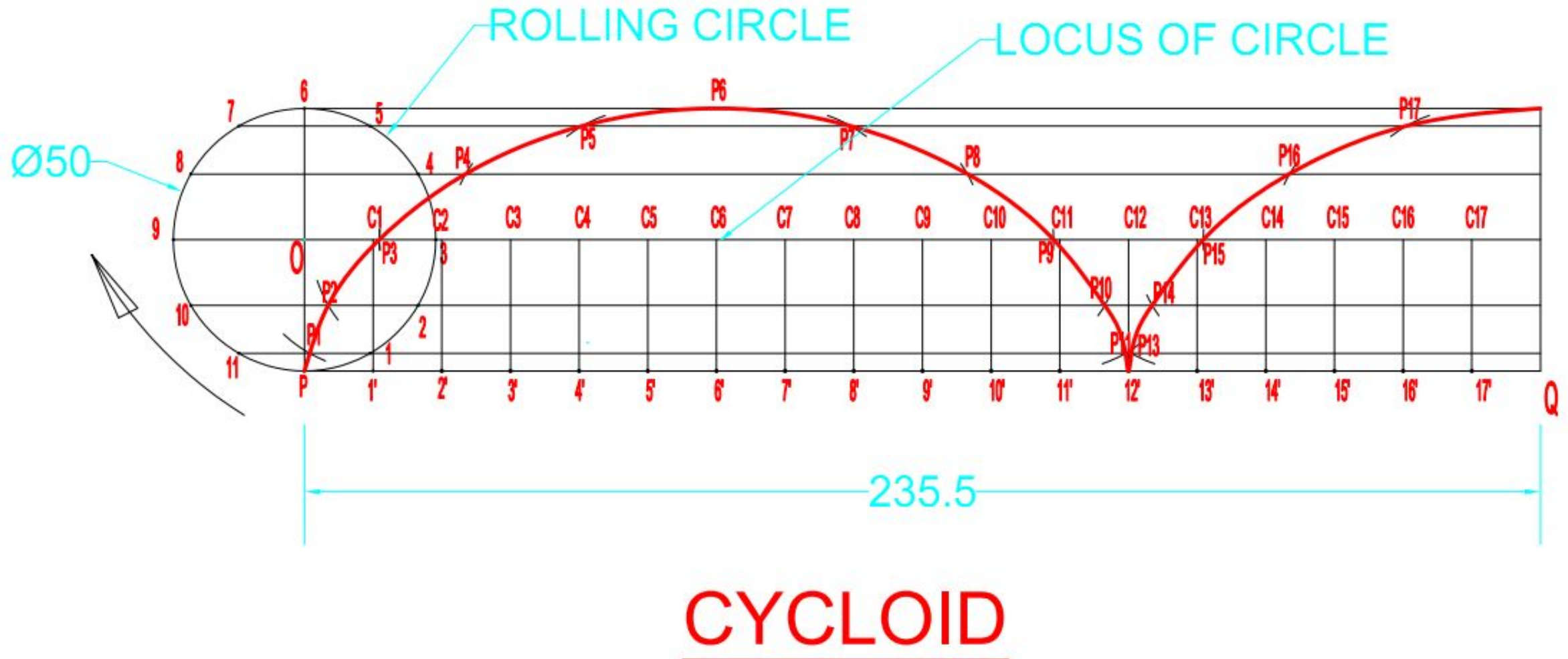
Exercise 5-5

6. Take an arc equal to OP (radius of circle) and mark from C1 to get P1, C2 to get P2, C3 to get P3 and so on.



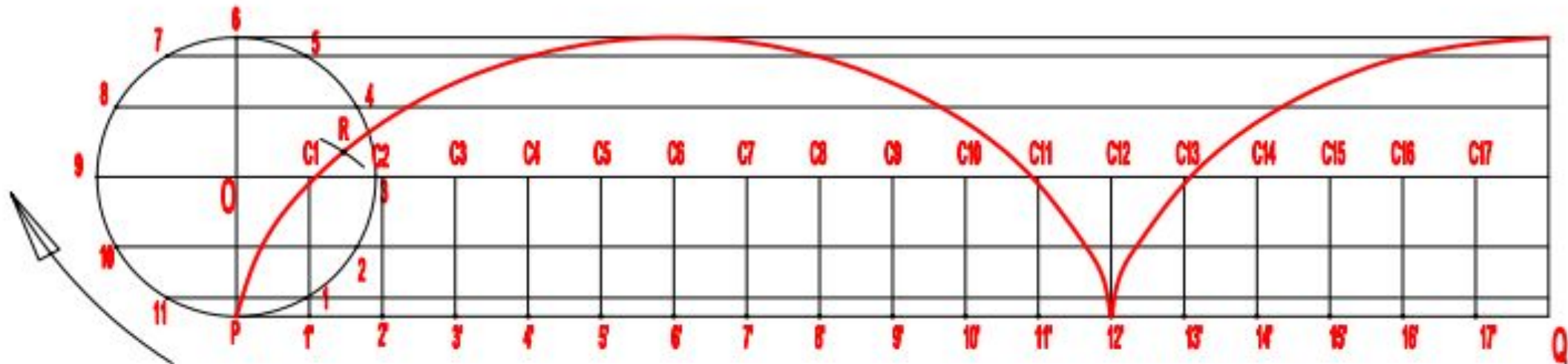
Exercise 5-5

7. Join the points to get a cycloid.



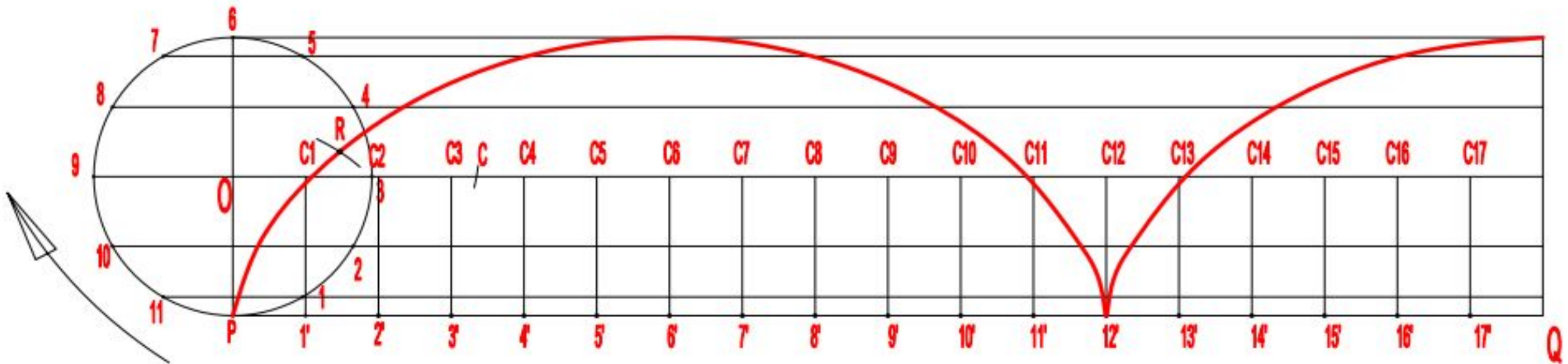
Exercise 5-5

8. For tangent, mark a length of 35 mm from on the curve as R.



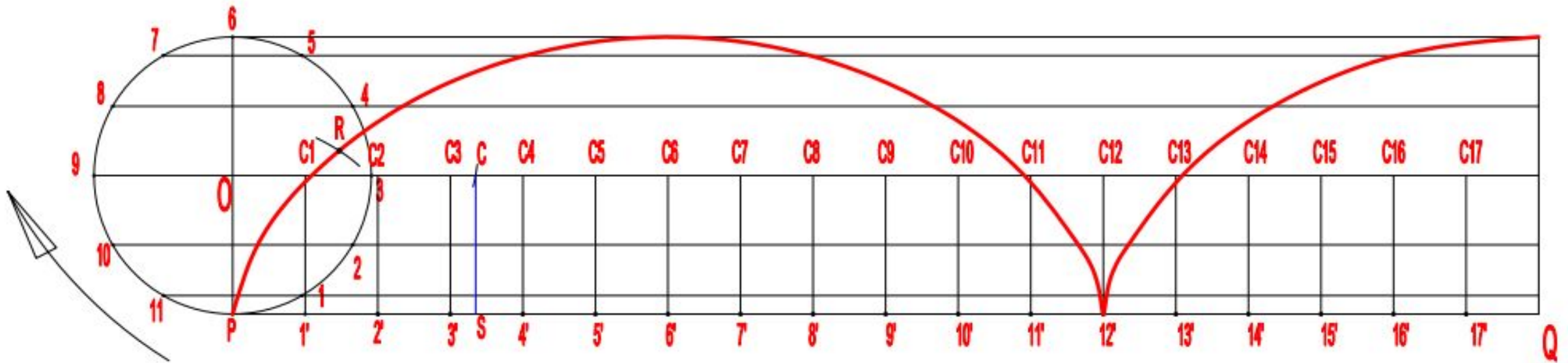
Exercise 5-5

9. Mark a length of OP from R as C as shown.



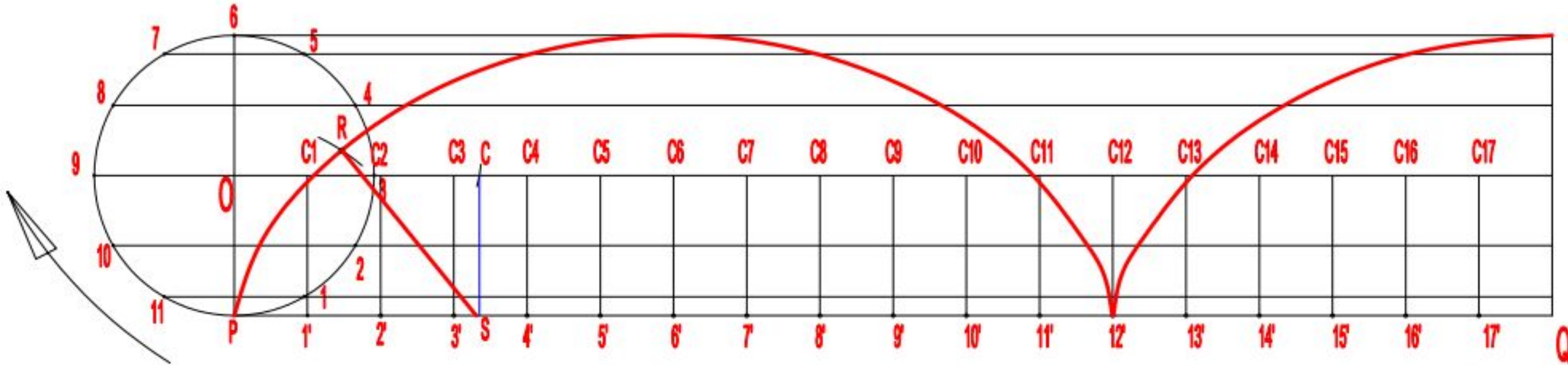
Exercise 5-5

10. Draw a vertical line from C to get S.



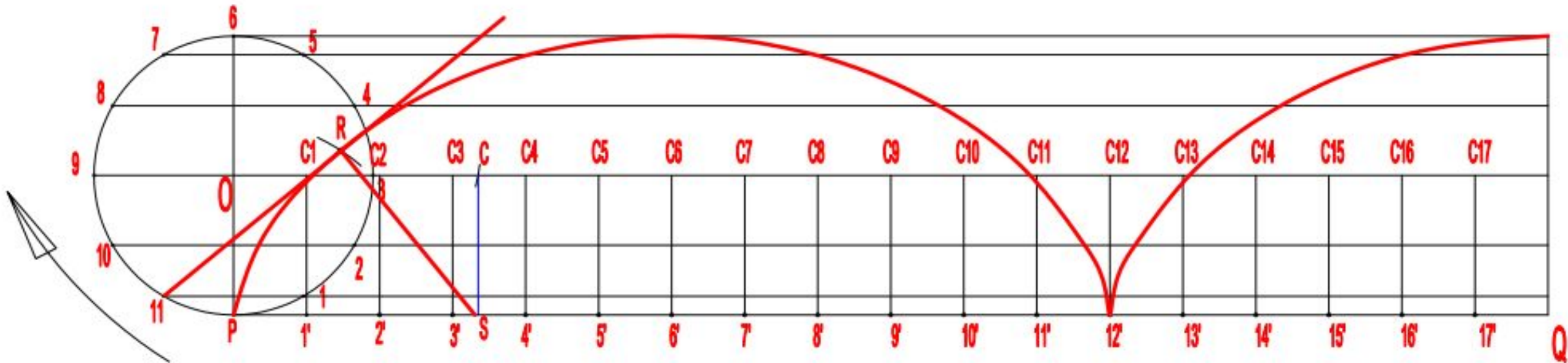
Exercise 5-5

11. Join R to S. This is the required normal.



Exercise 5-5

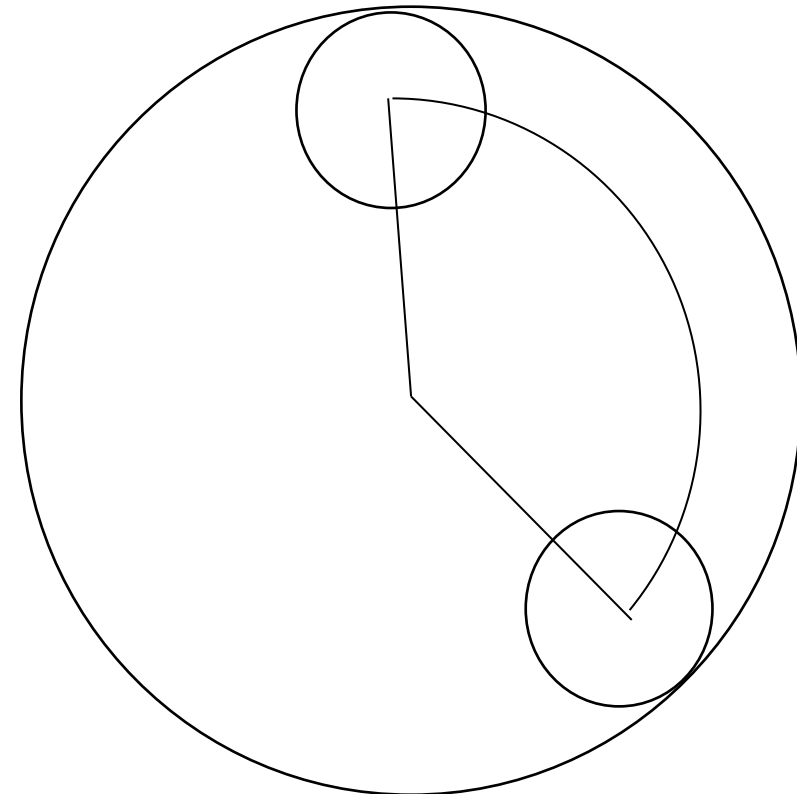
12. Draw a line perpendicular to RS through R. This is the required tangent.



Exercise 5-6 (Hypocycloid)

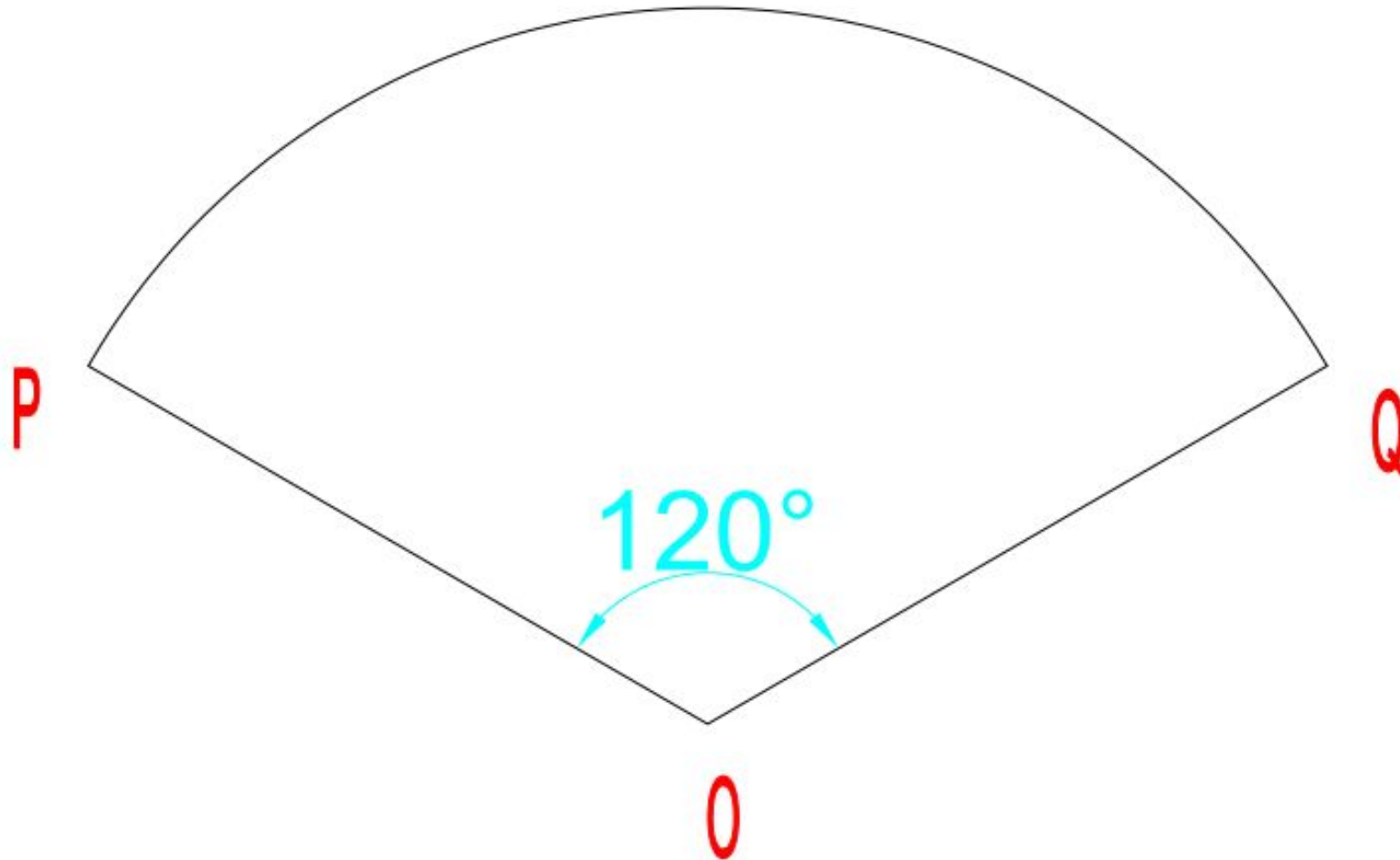
A circle of diameter 36mm rolls *inside* a circle of diameter 108mm. for one revolution without slipping. Trace the locus of a point on circumference of the circle of 36mm diameter. Name the curve. Also draw the tangent and normal at any point on the curve.

- Outer radius $R = 108/2 = 54$ mm
- Inner radius $r = 36/2 = 18$ mm
- Outer arc $2\pi R$ subtends an angle of 360° at the center O.
- Inner arc $2\pi r$ subtends an angle equal to $\frac{2\pi r}{2\pi R} \times 360^\circ$
i.e. 120° at O.



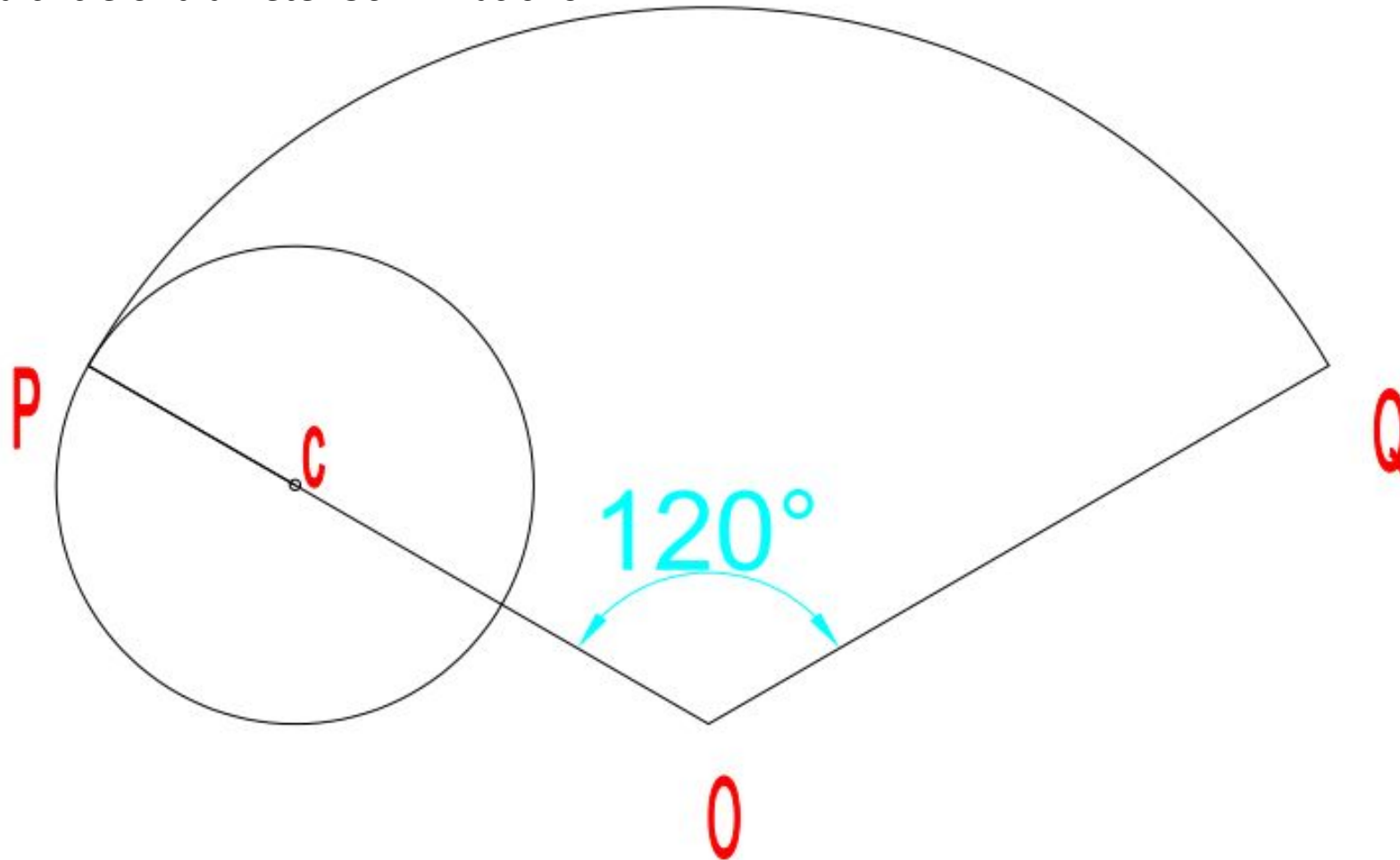
Exercise 5-6 (Hypocycloid)

1. Draw an arc of diameter 108 mm with center O.
2. Cut the arc to get OP and OQ such that Angle POQ = 120 degree.



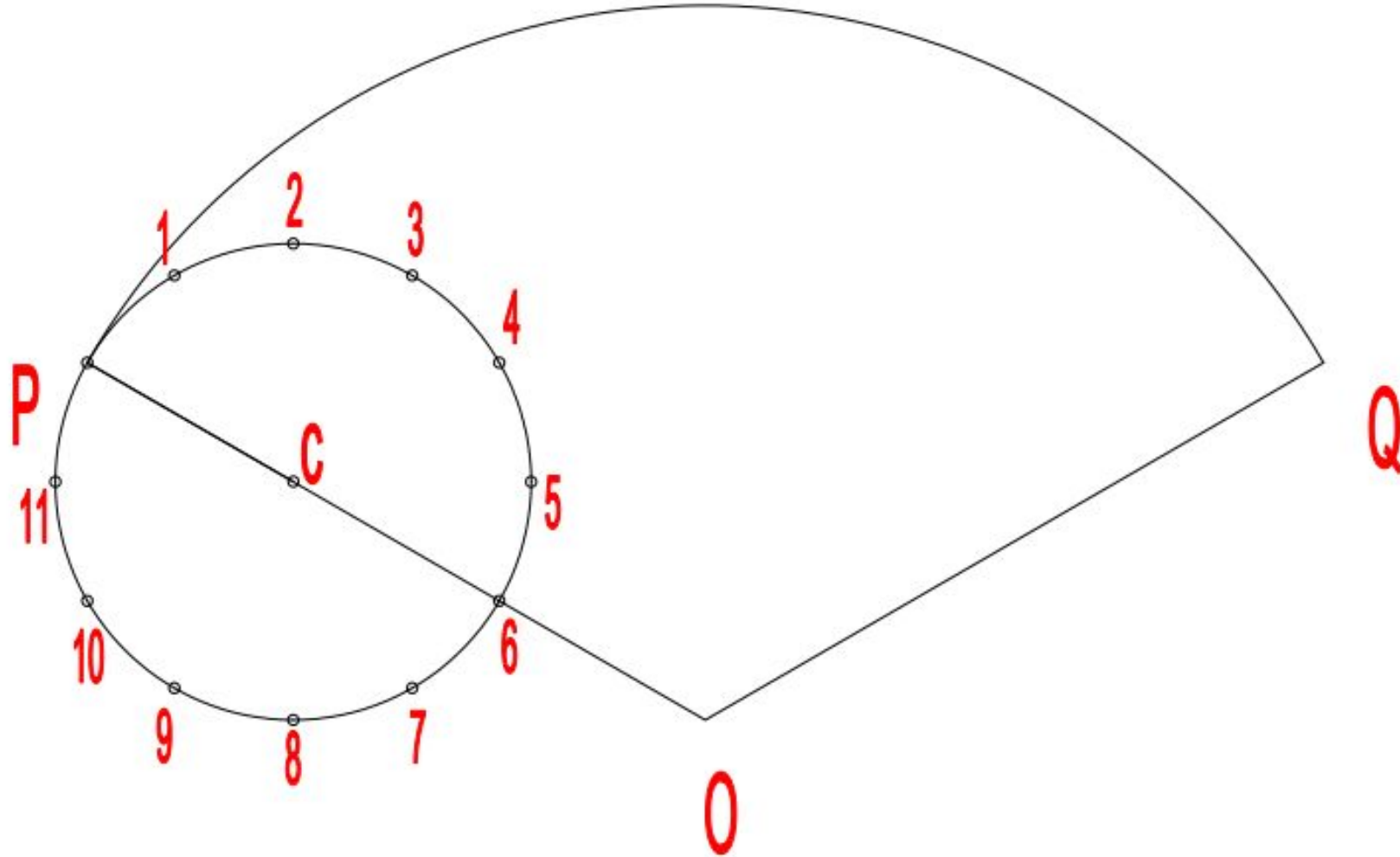
Exercise 5-6 (Hypocycloid)

3. Mark a point C as a distance of $36\text{mm}/2 = 18\text{ mm}$ from P along the line OP.
4. Draw a circle of diameter 36 mm as shown.



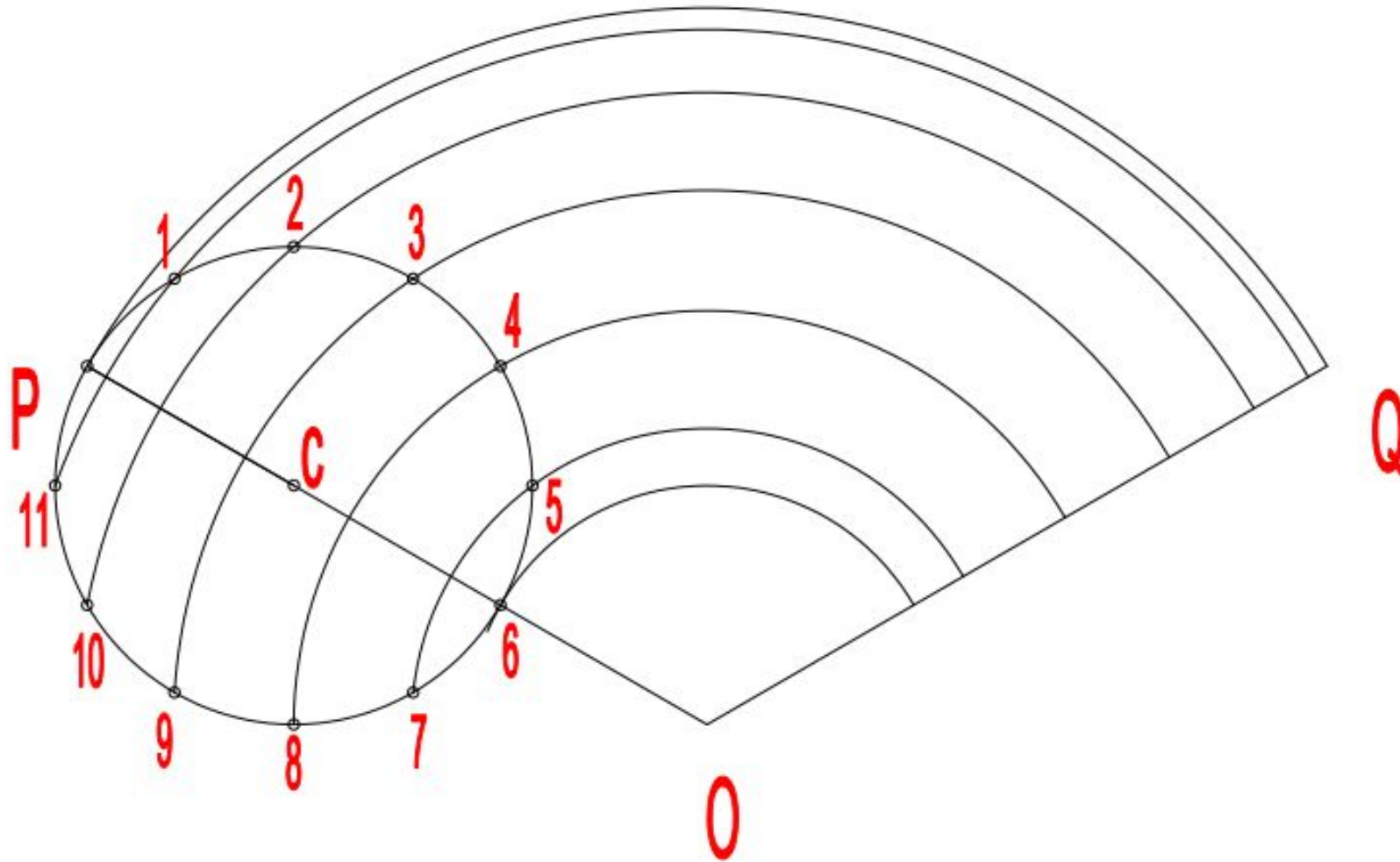
Exercise 5-6 (Hypocycloid)

5. Divide the smaller circle into 12 parts.



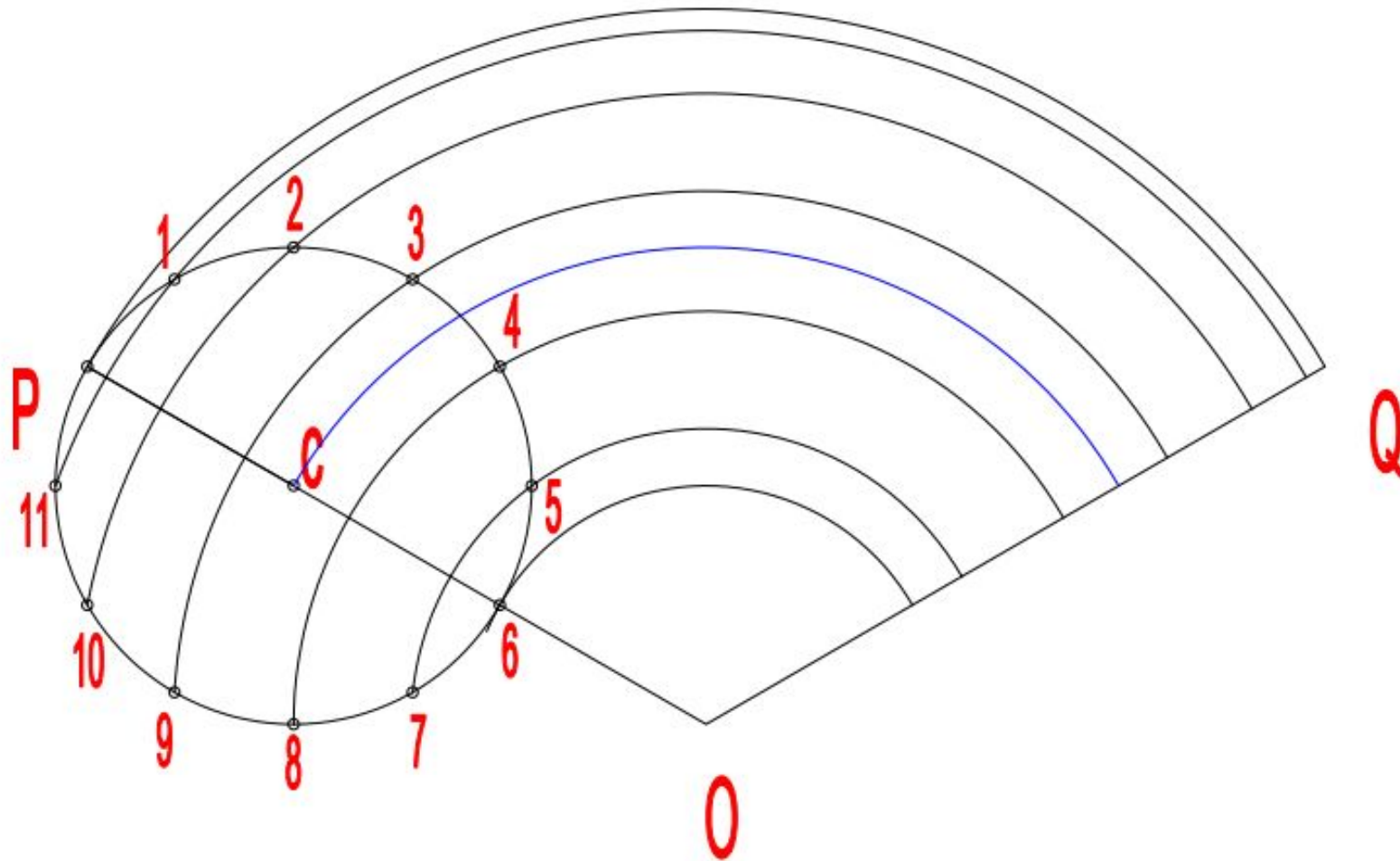
Exercise 5-6 (Hypocycloid)

6. Draw arcs from divisions of the smaller circle with center O as shown.



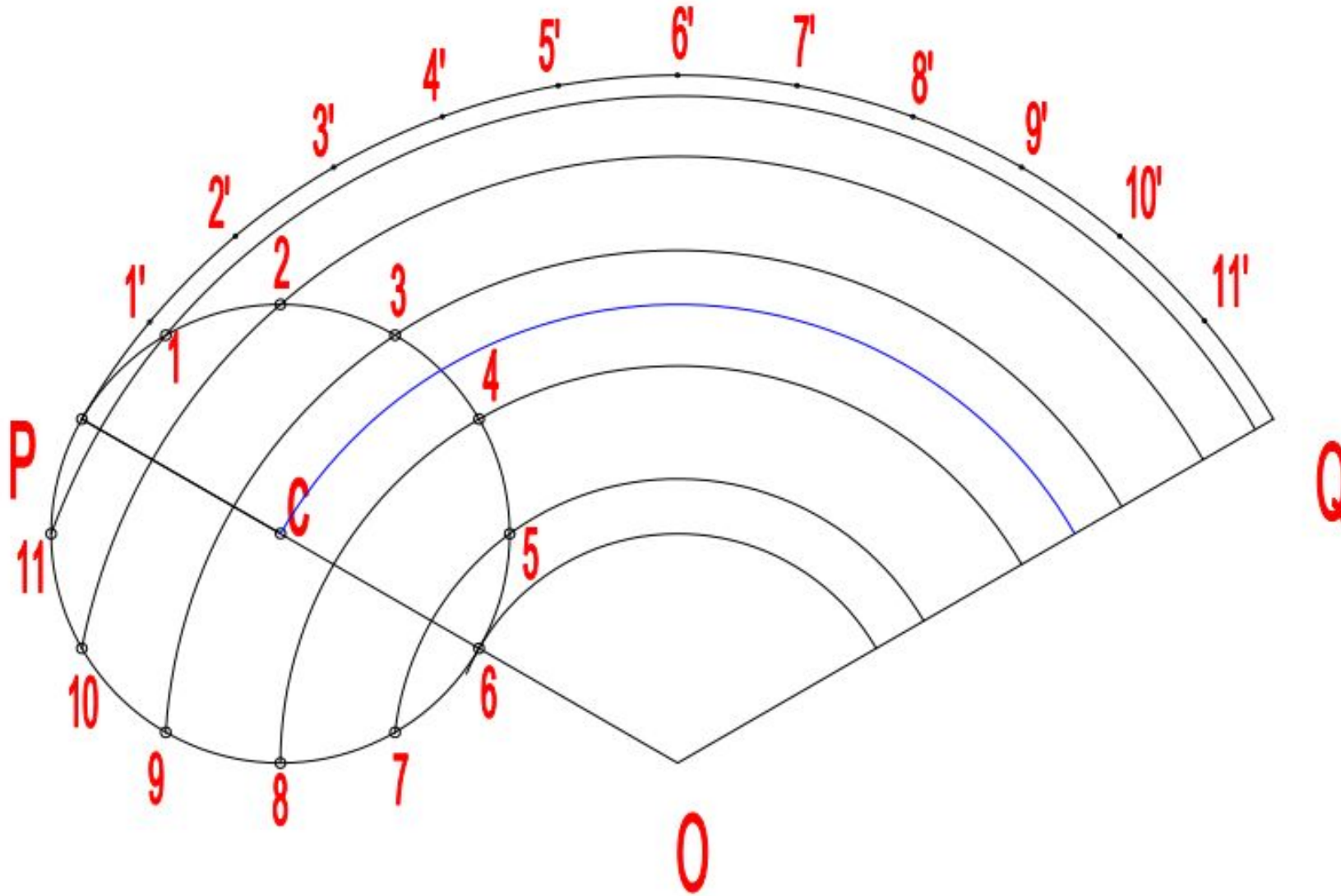
Exercise 5-6 (Hypocycloid)

7. Draw arc from C as shown.



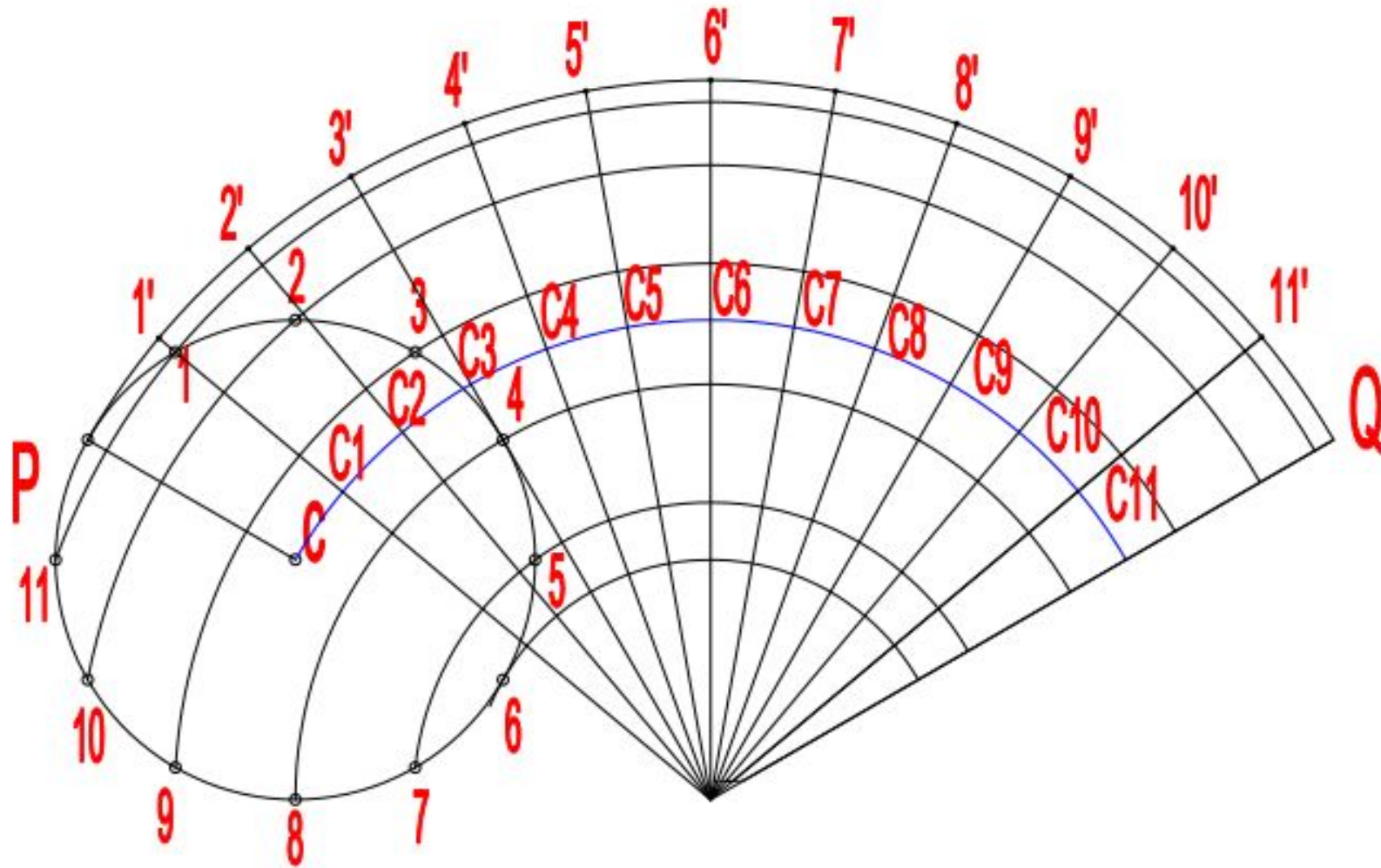
Exercise 5-6 (Hypocycloid)

8. Divide the bigger arc into 12 equal divisions. (for 1 revolution)



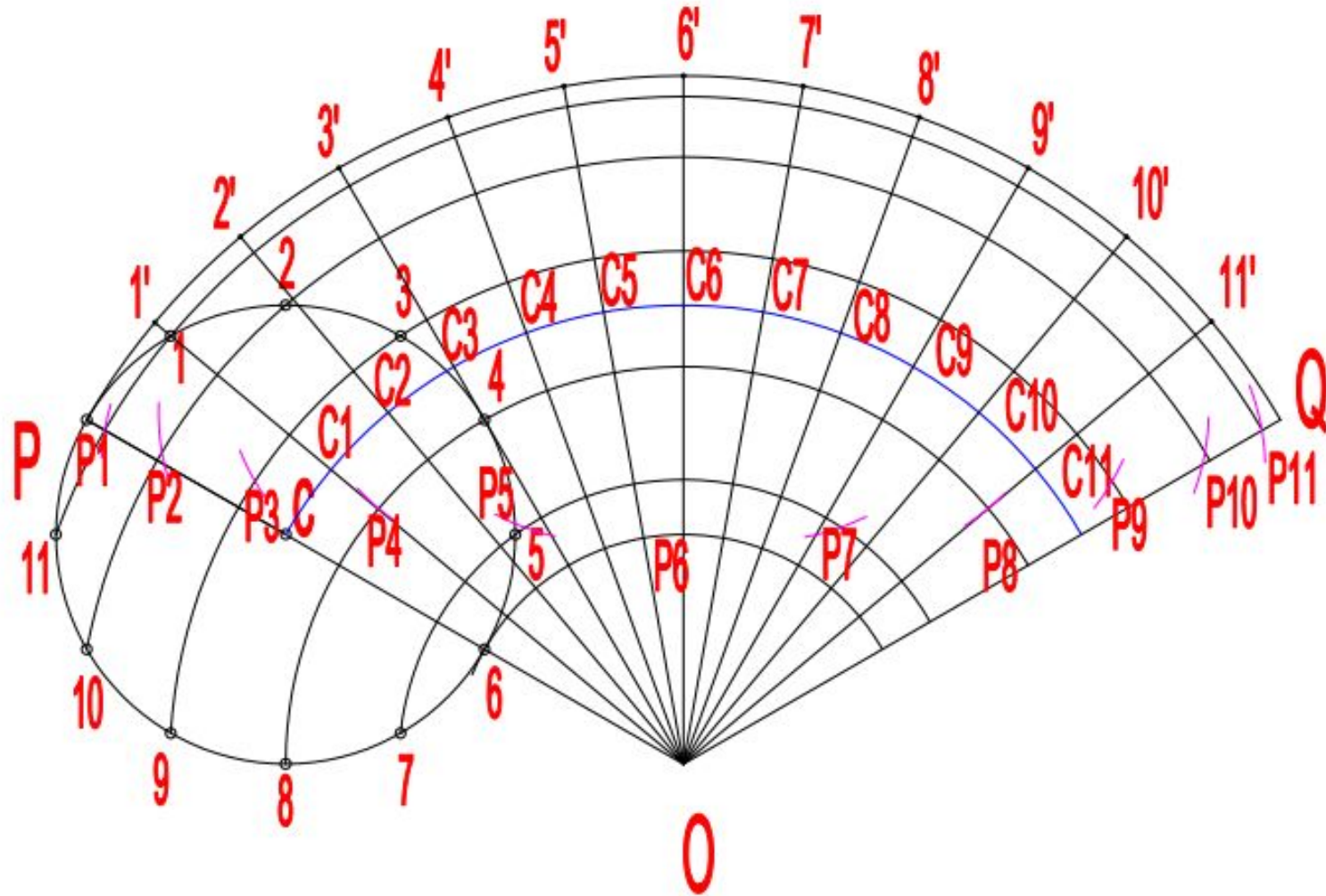
Exercise 5-6 (Hypocycloid)

9. Join each division of the arc with O to get locus of circle as C1, C2, C3, etc.



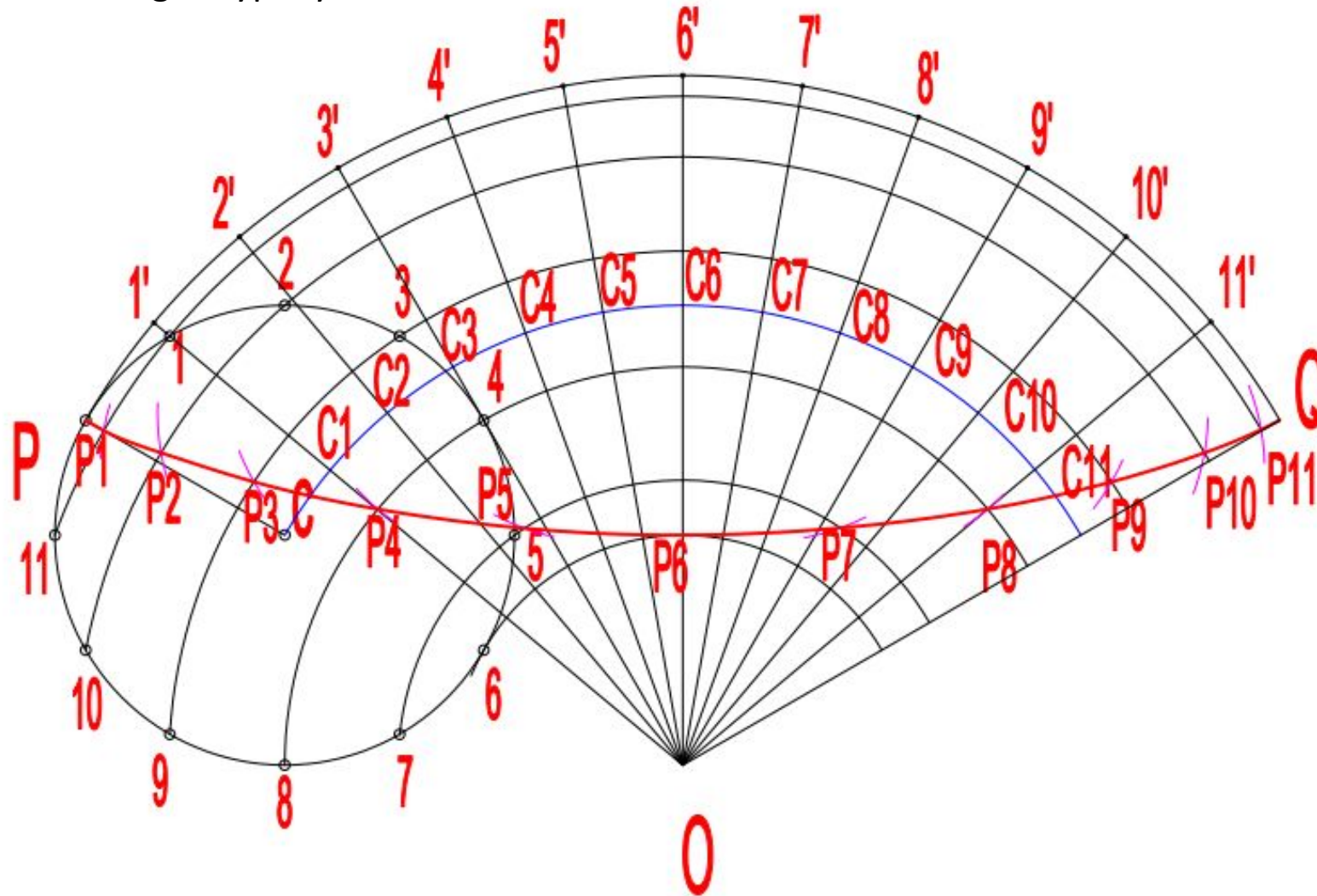
Exercise 5-6 (Hypocycloid)

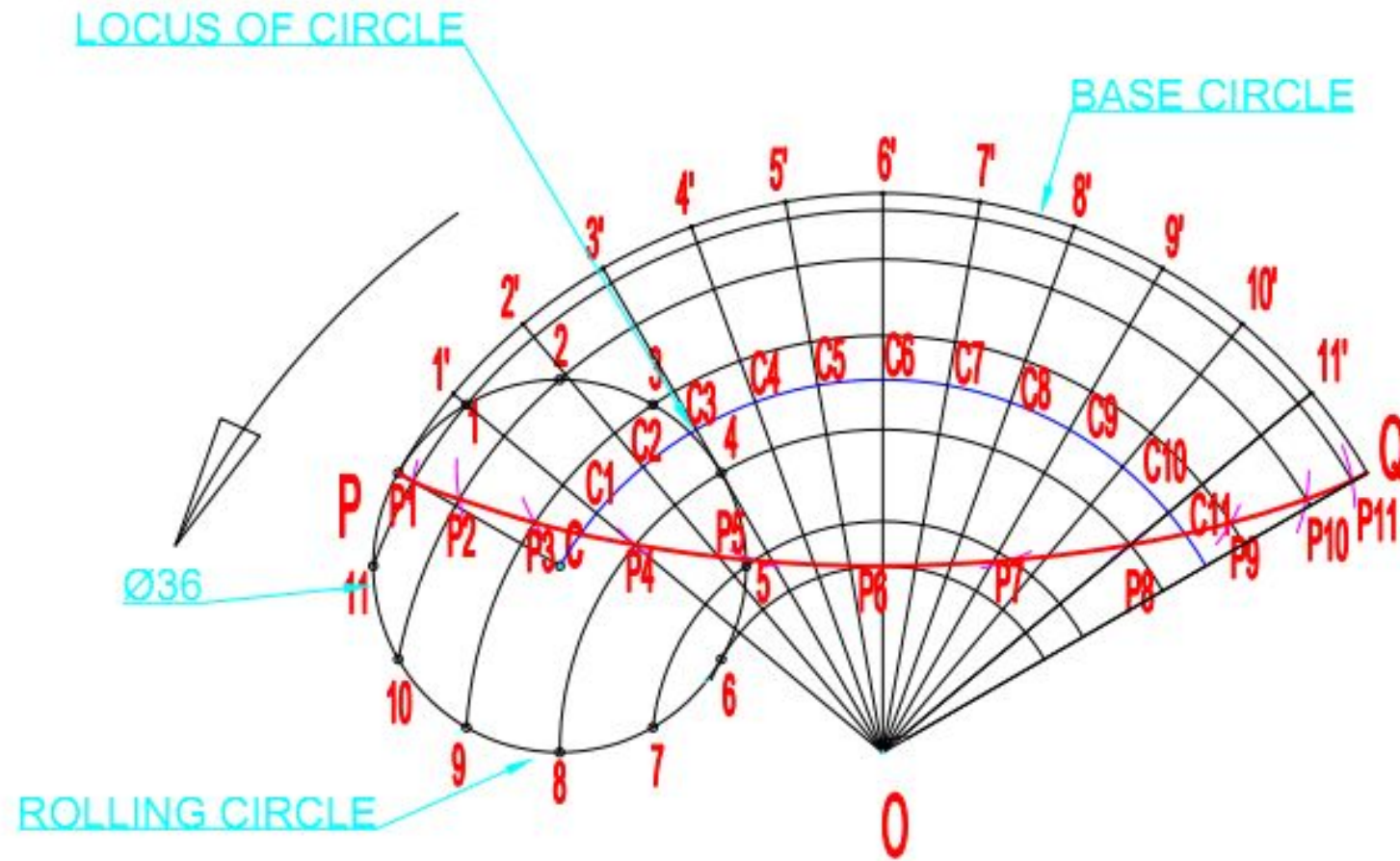
10. Take arc equal to CP (radius of smaller circle) and cut from C1 to get P1, C2 to get P2, C3 to get P3, etc. as shown.



Exercise 5-6 (Hypocycloid)

11. Join the arcs to get hypocycloid.





HYPOCYCLOID

Dividing an arc into equal parts

1. Draw the arc and extend it to get a 90 degree – arc.
2. Divide a line into 8 equal parts as shown.
3. Divide the 90 degree – arc into 8 equal parts by the method of bisecting.
4. Extend the divisions of this arc upto the divisions of the line as shown.
5. Get the sine wave.
6. Now extend end points of the arc, which has to be divided, horizontally upto the sine wave and then proceed downwards vertically.
7. Now divide this part of line into the number of divisions you want to divide the arc into.
8. Extend these divisions upwards vertically upto sine wave then proceed horizontally upto the arc.
9. Thus the arc is divided.

